



Review

A comprehensive review of solid-state lithium batteries: Fast Charging characteristics and in-operando diagnostics

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ABSTRACT

This comprehensive review article delves into the evolving landscape of solid-state batteries (SSBs), presenting a critical evaluation beyond the conventional lithium-ion technology. It meticulously explores the performance analysis, fast-charging capabilities, and in-operando characterization of SSBs. Amid growing demands for safer, higher energy density, and more sustainable energy storage solutions, SSBs emerge as a compelling alternative, promising to revolutionize applications from portable electronics to electric vehicles. This review begins with a detailed examination of the fundamental principles underpinning SSBs, including their unique architecture and the roles of solid electrolytes in enhancing safety and energy density. It then shifts focus to the pivotal aspect of fast charging – a critical parameter for consumer acceptance and market penetration of electric vehicles. The analysis extends to the intricacies of in-operando characterization techniques, which are essential for understanding the dynamic processes within SSBs during operation, thus offering insights into optimizing performance and longevity. By juxtaposing SSBs with conventional lithium-ion batteries, the review highlights the technological strides and hurdles ahead. Furthermore, it discusses the latest advancements in materials science, engineering strategies, and characterization methods that potentially address the current limitations of SSBs. Conclusively, this review not only underscores the significant progress in SSB technology but also outlines the roadmap and future directions for researchers and industry stakeholders aiming to harness the full potential of SSBs in next-generation energy storage systems.

1. Introduction

In solid-state batteries (SSBs), solid electrodes and electrolytes are used in replacement of typical liquid or polymer gel electrolytes of conventional batteries (i.e., lithium-ion batteries – LIBs). This solid-state technology has existed for quite some time and is proposed to solve many issues presented by the use of liquid electrolytes in batteries [1]. Therefore, in the past two decades, much research has been targeted at improving SSBs for ubiquitous use in critical technologies, such as electric vehicles (EVs). Liquid electrolyte batteries were projected to reach practical limits of energy density given the predicted constraints of materials technology development [2–5], making them less than favorable for use in energetically demanding technologies such as EVs. Moreover, SSBs pose numerous advantages over LIBs, including

increased energy storage density, life-cycle, charging speed, and increased safety due to higher thermal stability (reducing flammability and volatility). In fact, Dudney et. al [6] predicted that SSBs would soon have the potential to be widely used in applications such as energy storage, as well for powering electronics devices and automobiles in 2015. While we have not reached this point, the evolution of SSBs is ongoing and promising; eight years later, with the continued global focus on decreasing fossil fuel dependency, we see major efforts toward applying solid-state technology. Particularly in the automobile industry, leading companies and research labs work toward implementing SSBs in EVs [7]. Nonetheless, much progress is still to be made, as many issues challenge the scale-up of production and actual advantageousness of SSBs over conventional batteries [8]. Specifically, performance enhancements that will make SSBs more cost effective per kWh, such as

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increasing power density for more efficient charging, as well as increasing life-cycle by solving mechanical issues, are key to the realization of SSBs in modern technology. Herein, we compare performance and cost of SSBs to liquid electrolyte batteries, as well as general challenges to implementation, then report on what is being done to improve SSBs.

High volumetric and gravimetric energy densities are important in determining the competitiveness of batteries in EV applications and, currently, respective densities of 770 Wh L^{-1} and 260 Wh kg^{-1} are reported for high-performing liquid electrolyte batteries. SSB volumetric energy densities are reportedly around 430 Wh L^{-1} , or 63 % of that of liquid electrolyte batteries, thus there is room for improvement [9]. Potential avenues for increasing these energy densities are explored in the “Future Prospects” section. Developments are promising though, as one startup – the Qing Tao Energy Development Co. – claims successful production of solid-state lithium batteries with gravimetric energy density of 400 Wh/kg with their 100 MWh per year production plant (as of 2019), which they begun expanding to 10 GWh as of March 2022 to increase production [10]. The United States Advanced Battery Consortium (USABC) reports that an 80 % charge reached within 15 min is necessary for EVs to be considered competitive against conventional combustion engine vehicles [11]. LIBs implemented in EVs, (i.e., Porsche Taycan, Tesla Model 3) reportedly reach 80 % charging in 23–27 min, while SSBs are claimed to meet the USABC requirement, charging within just 15 min. Moreover, LIBs show degradation in just 1000 cycles, while SSBs are more durable – in 5000 cycles, they are able to retain 90 % of the original battery capacity, due to lessened mechanical strain. Finally, a major performance advantage of SSBs over LIBs is their avoidance of liquid electrolytes and carbon anodes, which not only limit battery energy density, but are also highly flammable [12–15]. Thus, use of the liquid electrolyte poses safety risks, arguably the most important consideration for EVs [16–18]. SSBs inherently address safety concerns in many ways, as shown in Fig. 1.

Theoretically, SSBs offer cost advantages over conventional liquid electrolyte batteries, a large component of their attractiveness. For instance, infiltration, formation, and conventional production steps for liquid electrolyte batteries are not required [3]. Such factors, in addition to significantly lower materials costs, ultimately work to make SSBs very cost effective. To this end, sulfide-electrolyte SSBs, designed for 6

GWh/year production output, are reportedly most cost-effective [19]. However, challenges of implementation specifically target the economical advantageousness of SSBs, and thus further exploration is required to actually realize the cost advantages. First, the materials used are an area of concern. Studies report that scale-up efforts are particularly prone to failure if the supply chain is constrained, as well as if cell integration warrants costly and impractical materials manufacturing [19]. Additionally, conventional processing and manufacturing techniques will need to be uniquely tailored to SSBs’ specific requirements, and integrating new procedures may present high upfront costs to further challenge implementation. Specifically, SSBs are reactive to humidity and oxygen, must be manufactured with high-pressures and temperatures, and possess unique mechanical properties which may pose manufacturability and safety-related complications [9]. Additionally, pertinent areas must be to be addressed as large-scale implementations ramp up, such as underdeveloped efforts to curb toxicity and make SSB technologies more sustainable (recyclable). Finally, defects and stability issues are prevalent and make SSB implementation imperfect [20]. Mainly, dendrite formation, which causes short-circuiting, is a problem that was expected to be curbed with use of solid rather than liquid electrolytes but is found to persist and is being extensively studied. Other mechanical failure mechanisms which threaten effectiveness and lifecycle life of SSBs include electrolyte fracture toughness, bending, and cracking. Fig. 2

As mentioned, SSBs were formerly assumed to be inherently resistant to dendrite formation compared to liquid electrolyte batteries, and although the mechanisms of dendrite growth are not necessarily similar between the two batteries, dendrites indeed persist in implementation of SSBs. This does not mean that the issue is as elusive as in LIBs [21–23]. In SSBs, dendrite formation is reportedly more predictable, thus can be managed through mindful materials engineering. Specifically, dendrites are shown to form when pores in the solid electrolyte are “inter-connected”; therefore, this issue may be combated by increasing the density of the electrolyte by altering the grain boundaries of the material, such as by sintering [24]. High-temperature sintering generally leads to a decrease in interfacial impedance or grain boundary resistance in ceramic materials. This reduction is primarily attributed to microstructural changes such as increased grain size and decreased porosity. As the sintering temperature rises, grain growth is enhanced, leading to

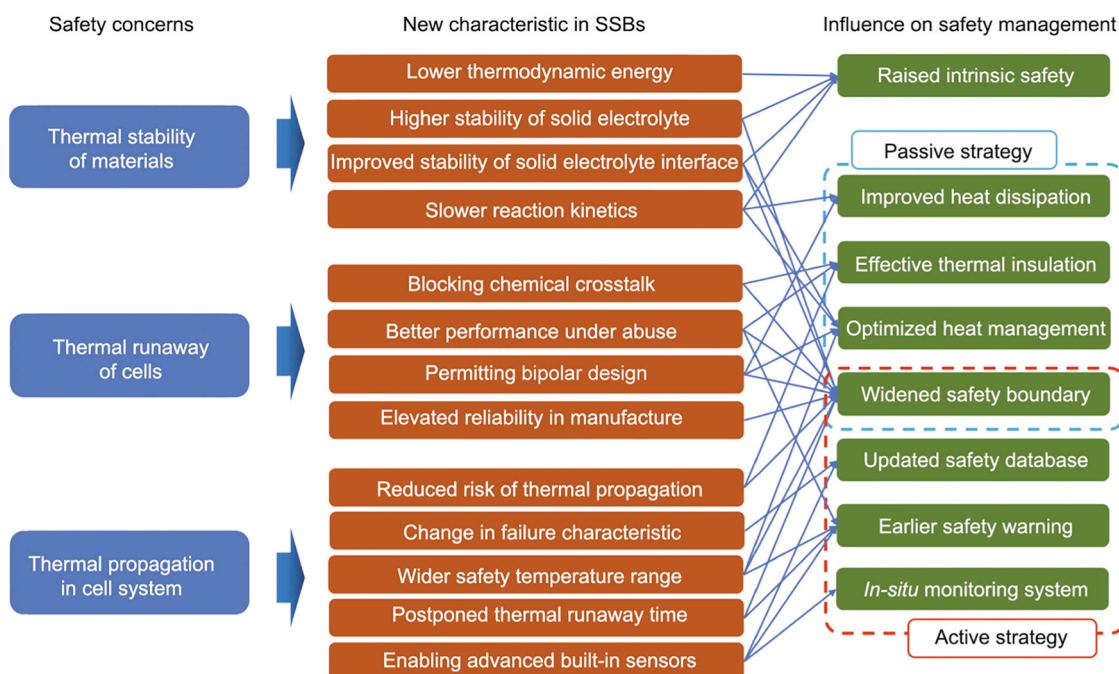


Fig. 1. Modifications in material and battery design across material, cell, and system tiers offer fresh avenues for enhancing safety strategies in SSBs [12].

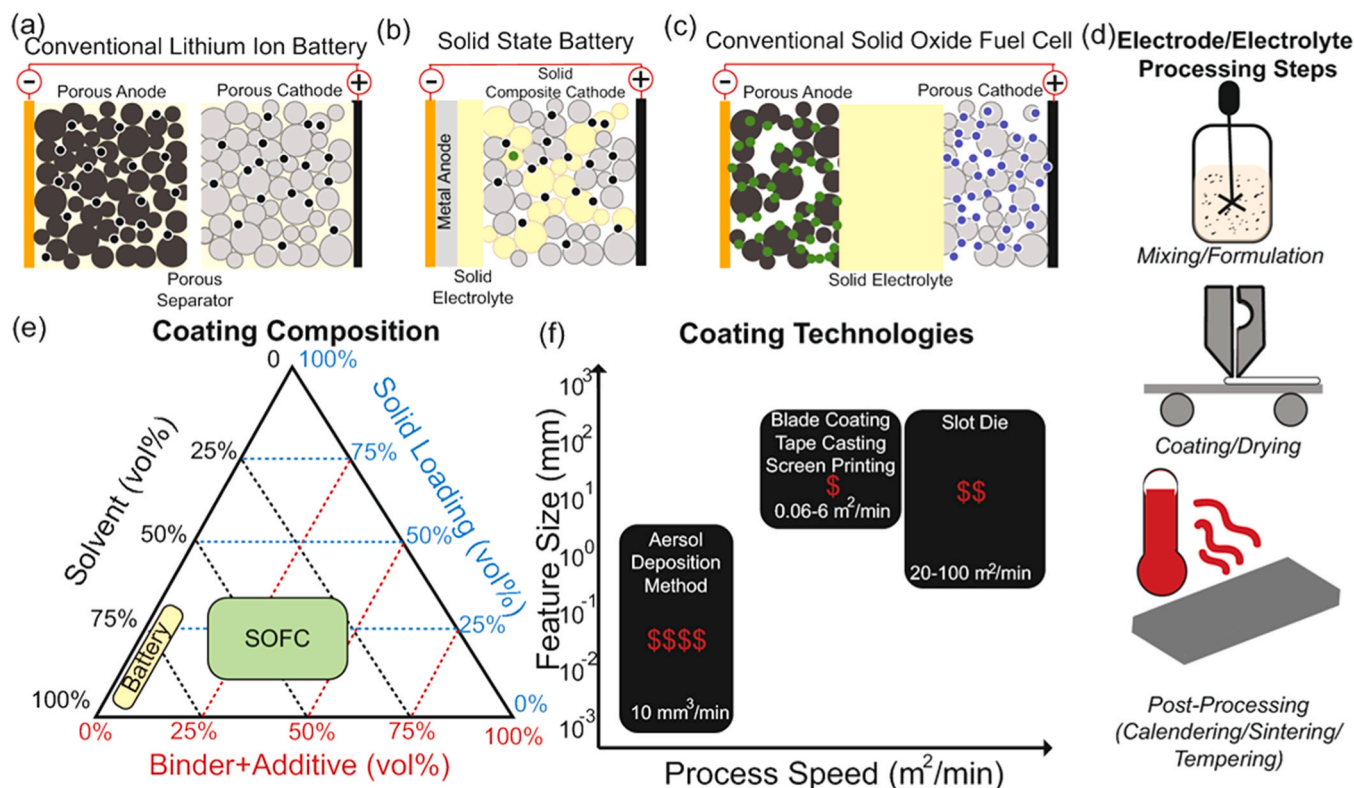


Fig. 2. Conventional Lithium-Ion battery and Solid-State battery diagrams [29]. (a) the basic composition of a conventional lithium-ion battery. (b) The composition of a solid-state battery. The major differences are that conventional batteries contain a porous anode and cathode while the solid-state battery consists of a metal anode, solid electrolyte, and solid composite cathode. (c) The conventional solid oxide fuel cell is similar to the conventional lithium battery with the exception of the solid electrolyte. (d) The processing steps for an electrode/electrolyte, which utilizes solution-based approaches for manufacturing. (e) The ink composition, which is engineered specifically for part performance. (f) The technologies used to process the inks.

fewer grain boundaries per unit volume, which are typically regions of higher resistance to ion or electron transport. Furthermore, higher temperatures promote densification by reducing the number and size of pores, improving the contact between grains and thus lowering the interfacial impedance. However, excessively high sintering temperatures can sometimes lead to secondary phase formation, elemental volatilization, or abnormal grain growth, which may counter the beneficial effects and potentially increase interfacial impedance. Therefore, an optimal sintering temperature range exists for minimizing grain boundary resistance [25–27]. Other material options are also being researched. Wang et al. at MIT mixed ion-electronic conductor tubes were filled with lithium (anode), such that the tubes are coated and thus are protected from the electrolyte, preventing dendrite formation [28]. Furthermore, the tube's 3D honeycomb architecture permits expansion/contraction that occurs during the charge/discharge process to avoid cracking. Such research on the optimization of anode and cathode materials can also help improve the energy density of SSBs. Coating techniques, such as those explored by Wang et al. can further improve thermodynamic instabilities. Other ways to improve energy density are being explored with thinner electrolyte layers, which are also reported to improve the economic attractiveness of SSBs. Therefore, such improvements are multifaceted, and future work should continue to search for optimized materials, manufacturing methods, and life-time properties of SSBs, as this will not only make them more efficient, but also more cost effective in comparison to liquid electrolyte batteries.

2. Manufacturing steps for SSBs

2.1. Transition to SSBs

The increased efficiency, capacity, and safety features have ramped

up the industry efforts to transition from lithium-ion batteries to solid-state batteries [29]. The movement of ions is facilitated by a solid electrolyte in solid-state batteries. The transition to solid-state batteries could increase the safety of batteries and improve the driving range needed to widely adopt the use of electric vehicles. Solid-state batteries use inorganic solid-state conductors that are non-flammable or have higher resistance to temperature in comparison to lithium-ion batteries which use organic liquid electrolytes [30]. The reactions that take place between the liquid electrolyte and the electrodes also contribute to the degradation of the battery. The use of a thermodynamically stable inorganic material can mitigate the decomposition of the battery electrolyte. The lack of solid materials containing ionic conductivities similar to liquid electrolytes at room temperature has partly limited the development of solid-state batteries. Recently, several solid electrolytes have been reported to have an ionic conductivity that is similar or higher than liquid electrolytes. However, the high conductivity is often nullified by the high impedance present at the interface between the electrode and the solid electrolyte. The addition of a buffer layer between the solid electrolyte and electrode is one of the most common strategies that has been developed to reduce the impact of the reactions that occur at the interface.

Solid-state batteries have emerged as one of the more realistic advancements to follow-up on the progress of lithium-ion batteries [3]. Bipolar stacking in solid-state batteries make high energy and power densities possible. In addition, electrode cross-talk could be prevented with the use of solid electrolytes. Electrode cross-talk causes instability in lithium-ion batteries due to the chemical interactions between the dissolved active materials. Solid-state batteries have faster charging capabilities due to the high lithium-ion transference number of inorganic solid electrolytes. Solid Power, Samsung, Toyota, and Quantum-Scape have made significant progress in the development of solid-state

battery technology. The safe use of high voltage cathode materials is one of the main advantages of solid-state batteries [31]. In contrast, the limited electrochemical stability window of liquid electrolytes does not allow for the use of these cathode materials.

2.2. The manufacturing process for SSBs

A study conducted by Schnell et al. compares the production of lithium-ion cells with the manufacturing processes for sulfide based solid-state batteries [32]. The study shows that the processes for composite electrode production could be modified to fit solid-state battery applications. However, the inclusion of a lithium metal anode and the production of the solid electrolyte separator layer will need to be developed completely. Solid-state batteries use lithium metal as the anode material, which is enabled by the compact solid electrolyte capable of acting as a barrier for lithium dendrites. The lithium metal creates a dense lithium anode as opposed to the porous anode used in the conventional lithium-ion battery. In addition, a dense cathode composite is used instead of a porous cathode. The layers are serially stacked in a dense manner as shown in Fig. 3b. The serial stacking is capable of increasing the voltage of a battery cell and optimizing the packaging design. Solid-state batteries also do not require the use of flammable organic components, so a cooling system is not needed. This makes solid-state batteries a much safer alternative to the conventional lithium-ion batteries.

Further research has shown that one of the main limiting factors of solid-state batteries is low ionic conductivity of the solid electrolyte [33]. Another critical bottleneck is the need for solid-state battery components with stable interfaces, which includes the solid electrolyte, conductive additives, and active material [34]. These major limiting factors along with the other associated limitations need to be resolved in order to efficiently manufacture solid-state batteries. Recently, solid electrolytes with adequate ionic conductivities have been achieved; however, engineering solid/solid interfaces with low impedance continues to be a challenge. Reaching the ideal composition could theoretically require extensive trial and error studies, which are both costly and time consuming. As a result, computational models have been used in various systems to predict the stability of solid electrolytes.

High-capacity anodes with low potentials could help achieve specific energies comparable with those of lithium-ion batteries [3]. One of the

most important things when it comes to charging batteries is that the capacity is not affected. Therefore, it is imperative that the anodes used for solid-state batteries allow a reversible high-rate operation. Currently, a silicon-based anode or lithium metal anode are the most probable to be used for the application of solid-state batteries. Abundant silicon anodes with ultrahigh theoretical capacity (4200 mAh/g) are one of the most promising options due to their extensive commercial prospects [35].

One of the main goals of current battery development is to design reversible lithium metal anodes that have high-rate capability [36]. Lithium metal anodes are capable of increasing the energy density of batteries that use intercalation-type cathodes. Specifically, lithium metal has a high theoretical capacity of 3860 mAh/g [37]. Solid-state batteries that use lithium metal anodes in tandem with solid electrolyte separators are promising because they are capable of preventing dendritic growth of lithium in the cell.

The use of inorganic solid electrolytes for solid-state batteries is beneficial because they are potentially safer than conventional batteries [38]. The ionic conductivities and mechanical properties of solid electrolytes are crucial for energy density and cycle performance improvements. Solid electrolytes such as $\text{Li}_7\text{P}_3\text{S}_{11}$, $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, and Na_3PS_4 have reportedly shown high ionic conductivities, and in some cases, similar to those of liquid electrolytes. There are several key features that solid electrolytes must include such as high ionic conductivity, minimal electrical conductivity, and the compound must be compatible with the anode and cathode materials [39]. The cathode is capable of oxidizing the solid electrolyte if the highest occupied molecular orbital of the electrolyte is above that of the cathode. Inclusively, it is important that the electrolytes act solely as an ion-conducting solid and do not actively take part in the insertion and extraction process of lithium.

Hitachi Zosen Corporation has reportedly designed a solid-state battery using machine technology via a proprietary fabrication process [40]. The structure consists of laminated positive electrode, solid electrolyte, and negative electrode layers. The solid electrolyte acts as a separator that successfully prevents contact between the positive and negative electrode interfaces while enabling the flow of lithium ions. The proprietary battery uses a non-flammable solid electrolyte. The company performed a nail penetration test on the battery and it did not smoke, ignite or explode, whereas conventional batteries would have ignited. In addition, the battery has a wide operating temperature range because the electrolyte does not solidify at low temperatures or

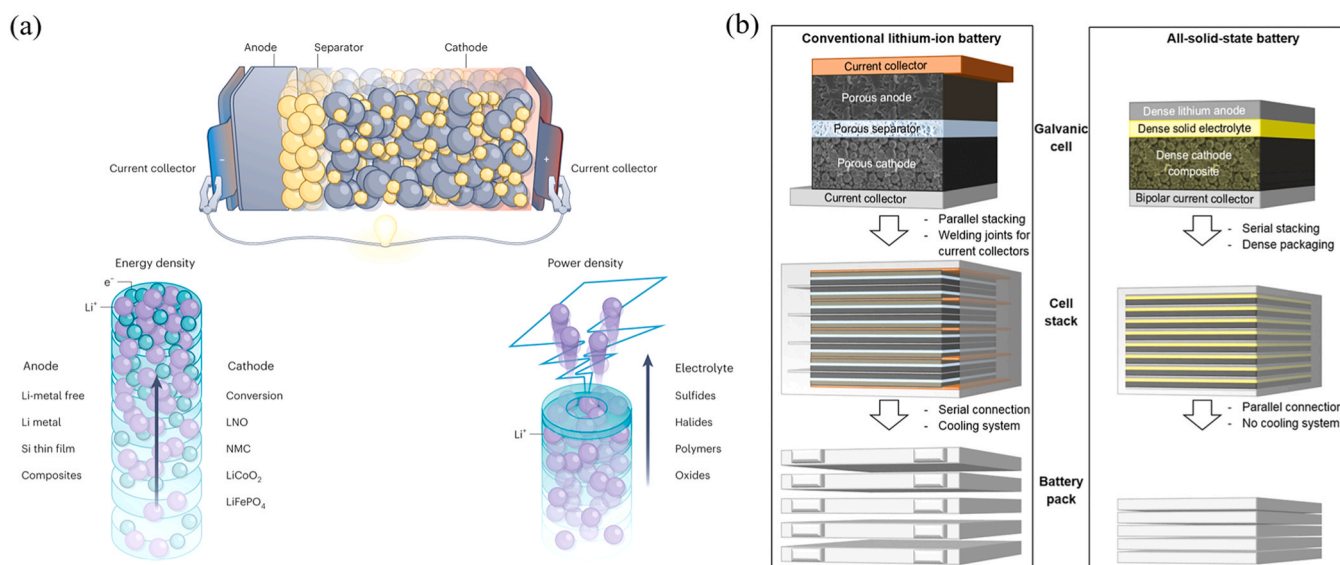


Fig. 3. (a) Visual representation of a lithium solid state battery cell [3]. this figure illustrates a conceptual representation of the lithium solid state battery cell. It also includes the most common anode, cathode, and solid electrolyte materials currently used. The anode and cathode materials on the left are listed in order of increasing energy density shown by the arrow. The electrolyte materials are listed in order of increasing power density for the drawing on the bottom right. (b) Composition comparison of the conventional Lithium-Ion battery and Solid-State battery [32].

decompose when exposed to high temperatures. The developed battery has an operating temperature range from -40°C to 120°C . It also contains a configuration that does not experience significant expansion even while under vacuum. The technology operates stably at a wide range of pressures, even in an environment of 1.0×10^{-2} Pa, and is currently being used selectively in aerospace applications, infrastructure, industrial applications and medical devices.

3. Performance analysis and observed challenges

While SSBs offer high energy density and safety, their fast-charging performance is currently limited by lower CCD and longer cycling times compared to LIBs. However, advancements in materials and interface design are promising for improving SSBs' fast-charging capabilities. LIBs, on the other hand, continue to lead in fast-charging performance due to better ionic conductivity and optimized charging protocols. Table 1 summarizes the comparative fast-charging performance of SSBs and LIBs.

3.1. Ionic and electronic conductivity

The ionic conductivity of the electrolyte directly impacts the battery's capacity, power density, and cycling stability. High ionic conductivity allows for fast transport of ions within the electrolyte, and high-power output and rapid charging and discharging rates. It also reduces the polarization and resistive losses at the electrode-electrolyte interfaces, improving the efficiency. High ionic conductivity allows for a larger active surface area of the electrode materials, leading to higher capacity and energy density. Low ionic conductivity can lead to poor battery performance, including low power output, slow charging and discharging rates, and low energy density. This can also result in electrolyte decomposition and the formation of unwanted interfacial layers, leading to capacity fading and reduced cycle life. The differences in ionic conductivity observed between halide electrolytes like Li_3YCl_6 and Li_3YBr_6 primarily stem from the size and polarizability of the halide ions, which influence the lithium ion transport pathways and energy barriers within their crystal structures. Bromide ions (Br^-) are larger and more polarizable than chloride ions (Cl^-) [50]. This larger size generally leads to an expansion of the lattice parameters, creating wider pathways for Li^+ ion migration and potentially lowering the activation energy for hopping between sites. The increased polarizability of Br^- can also affect the electrostatic interactions within the lattice, further influencing Li^+ mobility. Consequently, Li_3YBr_6 often exhibits higher ionic conductivity compared to Li_3YCl_6 at the same temperature [50]. However, this substitution can also impact other properties like electrochemical stability. Besides, the densification technique significantly impacts both the mechanical strength and ionic conductivity of sulfide-based SSBs. Hot pressing, which involves applying high pressure at elevated

temperatures, generally leads to higher density, enhanced grain connectivity, and reduced porosity in the solid electrolyte. This improved microstructure typically results in higher mechanical strength and enhanced ionic conductivity due to a larger number of conducting pathways and reduced grain boundary resistance. In contrast, cold sintering, performed at lower temperatures (often room temperature) with pressure, may result in lower density and potentially higher porosity compared to hot pressing. While cold sintering can still yield acceptable ionic conductivity, the mechanical strength might be lower due to weaker inter-particle bonding. However, cold pressing is often employed for sulfide electrolytes due to their inherent softness and ability to achieve sufficient density without high temperatures, which can be detrimental to the electrolyte's stability or interface formation with electrodes. The choice of densification technique must therefore balance the need for high mechanical integrity and ionic conductivity with the processing limitations and stability of the specific sulfide material and overall battery design [51].

Electronic conductivity has a crucial role in the cathode and anode materials of SSBs. The transport of electrons occurs at the electrode-electrolyte interfaces, which determines how efficiently electrons can be transported. High electronic conductivity of the electrode leads to high interfacial resistance which means poor battery performance, and causes issues such as low power output, voltage polarization, and capacity loss. Low electronic conductivity can limit the rate of electrochemical reactions and lead to slow charging and discharging rates. Various strategies to improve the electronic conductivity of electrode materials, such as the use of conductive additives or coatings, such as carbon nanotubes, graphene, or metallic nanoparticles, can improve the electronic conductivity of the electrode materials. Other strategies also include doping the electrode materials with elements that can enhance electronic conductivity, such as transition metals or rare earth elements. To decouple and quantify interfacial phenomena in practical solid-state battery architectures, several experimental techniques are proposed. Electrochemical Impedance Spectroscopy (EIS) is a widely used technique to separate and quantify different resistive components within the battery, including interfacial resistances arising from charge transfer and ion migration at the electrode-electrolyte interfaces. By analyzing the impedance response over a range of frequencies, individual interfacial contributions can be deconvoluted. In-situ and operando characterization techniques such as X-ray Diffraction (XRD), Transmission Electron Microscopy (TEM), Scanning Electron Microscopy (SEM), and X-ray Photoelectron Spectroscopy (XPS) allow for the real-time monitoring of structural and chemical changes at the interfaces during battery operation. These methods can help identify the formation of secondary phases, interfacial voids, or changes in chemical states that contribute to interfacial impedance. Kelvin Probe Force Microscopy (KPFM) can be employed to image the local potential variations at the interfaces, providing insights into the potential barriers that impede ion transport. Furthermore, thermal wave sensing using modified 3ω sensors offers a non-invasive method to probe the morphology of the lithium metal-solid electrolyte interface and relate the thermal interface resistance to the physical structure [52,53]. Combining these techniques provides a comprehensive approach to decouple and quantify the complex interfacial phenomena in solid-state batteries. Tables 2–5. (Fig. 4)

Along with these forms of transport in ceramic solid electrolytes, conduction across grain boundaries can offer more resistance than intragrain conduction. Li^+ conductivity in ceramics can be described by the product of the number of mobile Li^+ per unit volume, the square of the charge of each ion, and the absolute mobility of the ions. The absolute mobility, μ , can be related to the diffusion coefficient D by the Nernst-Einstein equation: Fig. 5

$$\mu = \frac{D}{k_B T} \quad (1)$$

With T as the temperature in Kelvin and k_B as the Boltzmann

Table 1

The comparison of fast-charging performance between lithium SSBs and state-of-the-art LIBs.

Parameter	Solid-State Batteries (SSBs)	Lithium-Ion Batteries (LIBs)
Critical Current Density (CCD)	Lower CCD due to low ionic conductivity and poor interface contact [41–43]	Higher CCD with better ionic conductivity [11,44]
Cycling Counts	High cycling stability with specific materials (e.g., 95 % capacity retention at 5 C for 1000 cycles) [45]	Generally high but can degrade faster under high-rate conditions [44,46–48]
Cycling Current Density	Limited by slow Li^+ transport and interface issues [43,45]	Higher due to better material and interface optimization [11,49]
Cycling Time	Longer due to slower kinetics and interface challenges [41, 43]	Shorter with optimized fast-charging protocols [47,48]

Table 2
Existing solid electrolytes and their properties.

Solid Electrolytes	Examples	Ionic Conductivity (mS cm ⁻¹)	Electronic Conductivity (S cm ⁻¹)	Transference Number (tLi ⁺)	Transport Mechanisms
Ceramics	Thio-LISICON conductor	2.2 [54]		Greater than 0.999 [56]	Li-ion transport depends on the defects found in the ceramic. This includes many types of defects, such as point, line, planar, volume and electron defects. Point defects have the strongest effect on ion transport, and these typically consist of Frenkel defects and Schottky defects. Ionic diffusion consists of random walks along point defects [54]. Li-ion transport mechanisms are accomplished through vacancy mechanisms, interstitial mechanisms, and interstitial-substitutional exchange mechanisms [54]. (Fig. 4a). The mechanisms that affect transport and grain boundaries are still not fully understood, but it is known that there are three possible types of transport: (1) intragranular Li-ion transport within grains seen in Fig. 4b, (2) Li ⁺ ions migrate from grain to grain by crossing boundaries seen in Fig. 4c, (3) Li ⁺ ions migrate along grain boundaries seen in Fig. 4d [54].
	Li _{3.25} Ge _{0.25} P _{0.75} S ₄				
	LGPS	12 [54]		0.56 [57]	
	Li ₇ P ₃ S ₁₁ glass-ceramic	17 [54]		~1 [58]	
	Nanoporous β-Li ₃ PS ₄	0.16 [54]		~1 [59]	
	Li _{6+x} P _{1-x} Ge _x S ₅ I	18.4 ± 2.7 (upon sintering) [54]			
	Garnets	0.1–1 [54]		0.77 (Ta-doped garnet Li _{6.4} La ₃ Zr _{1.4} Ta _{0.6} O ₁₂) [60]	The mechanisms that affect transport and grain boundaries are still not fully understood, but it is known that there are three possible types of transport: (1) intragranular Li-ion transport within grains seen in Fig. 4b, (2) Li ⁺ ions migrate from grain to grain by crossing boundaries seen in Fig. 4c, (3) Li ⁺ ions migrate along grain boundaries seen in Fig. 4d [54].
	Thin-film LiPON	~10 ⁻³ [54]	Ranging from 10 ⁻¹² to 10 ⁻¹⁴ [54]	1 [61]	
	Perovskites	~1 [54]		10 ⁻⁶ ~ 1 [62]	
	Antiperovskites	Ranging widely from 10 ⁻⁴ to 1 [54]		0.84 [63]	
Halides	NASICONs	> 0.1 [54]		0.88 [64]	The transport mechanisms in halides are similar to those found in ceramics [54]. Segment motion is the common transportation mechanism found in polymer electrolytes. Ion transport is directly dependent on the segmental mobility of polymer chains, which often occurs in amorphous regions above the glass transition temperature (Fig. 4e) [54]. The mean free path of ion hopping is relatively large for polymers when compared to that in inorganic electrolytes, but more studies need to be done on the ion-hopping mechanism in polymers. It has been shown that polymer cement and ion motion can cooperatively happen, and Li ⁺ ions transport along the polymer chains before jumping from one chain to another [54]. For ceramic-polymer composites with inorganic fillers, Li ⁺ ions transport occurs via a mechanism similar to that of a solid polymer electrolyte. For composites with active fillers, the ion conduction mechanism is more complicated. There are three possible ion transport pathways: (1) intrapolymer transport, (2) intrac ceramic transport, and (3) interfacial polymer/ceramic transport [54]. The conduction pathways can also depend on the shape of the filler that is used, whether that is nanoparticles, random nanowires, or aligned nanowires.
	Li ₃ YCl ₆	0.51 [54]	2.8 × 10 ⁻⁹ [55]	1 [65]	
	Li ₃ YBr ₆	0.72 [54]	10 ⁻⁹ [55]	~1 [66]	
Polymers	Solid cross-linked polymers	0.24 [54]		0.53 ~ 0.63 [67]	Segment motion is the common transportation mechanism found in polymer electrolytes. Ion transport is directly dependent on the segmental mobility of polymer chains, which often occurs in amorphous regions above the glass transition temperature (Fig. 4e) [54]. The mean free path of ion hopping is relatively large for polymers when compared to that in inorganic electrolytes, but more studies need to be done on the ion-hopping mechanism in polymers. It has been shown that polymer cement and ion motion can cooperatively happen, and Li ⁺ ions transport along the polymer chains before jumping from one chain to another [54]. For ceramic-polymer composites with inorganic fillers, Li ⁺ ions transport occurs via a mechanism similar to that of a solid polymer electrolyte. For composites with active fillers, the ion conduction mechanism is more complicated. There are three possible ion transport pathways: (1) intrapolymer transport, (2) intrac ceramic transport, and (3) interfacial polymer/ceramic transport [54]. The conduction pathways can also depend on the shape of the filler that is used, whether that is nanoparticles, random nanowires, or aligned nanowires.
	PPO-PEO-PPO (PPO = poly[propylene oxide], PEO = poly[ethylene oxide])	1 [54]		0.2 ~ 0.4 [68]	
Composites	PEO-Li with SiO ₂ and EC-PC (passive filler)	0.2 [54]		0.26 ~ 0.42 [69,70]	For ceramic-polymer composites with inorganic fillers, Li ⁺ ions transport occurs via a mechanism similar to that of a solid polymer electrolyte. For composites with active fillers, the ion conduction mechanism is more complicated. There are three possible ion transport pathways: (1) intrapolymer transport, (2) intrac ceramic transport, and (3) interfacial polymer/ceramic transport [54]. The conduction pathways can also depend on the shape of the filler that is used, whether that is nanoparticles, random nanowires, or aligned nanowires.
	PEO-Li with MB-LLZTO (active filler)	0.311 [54]		0.3 ~ 0.77 [60,71]	

constant. Thus, the conductivity can be expressed as:

$$\sigma = \frac{\sigma_0}{T} \exp\left(\frac{-E_A}{k_B T}\right) \quad (2)$$

Where E_A is the activation energy of diffusion.

The ionic conductivity in solid electrolytes can be measured using impedance spectroscopy (AC technique), where a sinusoidal potential at various frequencies is applied to an electrochemical cell with blocking electrodes and the response is recorded. The resulting impedance results can be analyzed using a combination of three RC-circuits, which represent the bulk conduction, grain boundary-conduction, and diffusion due to concentration gradient in the electrode. The resistance from each portion of the circuit can be extracted, which allows for the ionic conductivity to be determined with a known thickness and area of the sample [75].

The electrical conductivity is measured by employing the Hebb-Wagner method, where one blocking and one non-blocking electrode is applied to the material, and measuring the current at steady-state as a function of applied potential. As one electrode is blocking, this leads to a buildup or depletion of mobile ions at the blocking electrode that is able to offset the applied electric potential [76]. Similar methods have been used to measure the ionic and electronic conductivity of halides. Polymer solid electrolytes utilize a different transport mechanism than

ceramic solid electrolytes. Ionic conductivity of polymer material under steady-state condition [77]:

$$\sigma = n_i q_i \mu_i \quad (3)$$

Where σ is the conductivity, n_i is concentration of the charge carrier, μ_i is the mobility, and q_i is the number of charges. For polymer electrolytes based on segment motion, the Vogel-Tamman-Fulcher equation is used to predict the temperature dependency of ion conductivity [54]. The equation states:

$$\sigma = \sigma_0 T^{1/2} \exp\left(-\frac{B}{T - T_0}\right) \quad (4)$$

Where σ_0 is the pre-exponential factor, B is the pseudo-activation energy for the conductivity, and T_0 is the equilibrium glass transition temperature. The table below shows some common techniques used to measure conductivity of polymer solid electrolytes.

Composite solid electrolytes composed of polymer matrix and inorganic fillers can improve on many performance parameters, such as ionic conductivity, mechanical properties, and thermal stability [41]. In a tri-phase composite electrolyte system consisting of inorganic nanofillers, interfacial region, and polymer/lithium salt, the overall ionic conductivity can be expressed as:

Table 3
Comparing existing techniques to measure conductivity of polymers [76].

Technique	Brief description/example	Advantages	Disadvantages
Four-probe on insulating substrates	Four electrodes attached to film on an insulating substrate; used for various applications such as OTFT's and free-standing films	Geometry ideal for measuring electronic conductivity of thin films	Electrode arrangement inappropriate for electrochemical doping because of using an insulating substrate; performed typically on dry film not in contact with liquid electrolyte; not suited for measurement of ionic conductivity
EIS with ion-blocking electrodes (Huggins approach)	Huggins approach of measuring the impedance of a material with two ion-blocking electrodes yielding ionic and electronic conductivity; P3HT-PEO block copolymer conductivity as a function of Li-salt mixing and ratio of PEO/P3HT	Simultaneously obtain ionic and electronic conductivity as a function of chemical doping	Electrode geometry inappropriate for electrochemical doping; thick polymer film needed if conductivity is high both electronic and ionic conductivities; can only be obtained when they are similar in magnitude
EIS + electrochemical doping	Huggins approach of measuring impedance with two ion-blocking electrodes; P3HT-PEO block copolymer electronic conductivity as a function of electrochemical doping using a 3-electrode cell	Obtain electronic conductivity as a function of electrochemical doping	Thick polymer films needed (>100 μm) for ensuring measurable resistance values for films with high conductivity values; typically used without liquid electrolyte; ionic conductivity only obtained when polymer is undoped otherwise shorted by electronic conductivity
EIS + interdigitated microelectrode (IDM)	EIS to obtain ionic conductivity of polymer thin films on IDM	IDM enhances signal; a thin film can be used; conductivity can be obtained as a function of chemical doping	Geometry inappropriate for electrochemical doping when either conductivity dominates; the other conductivity value cannot be determined; no liquid electrolyte can be used
Direct current transient method with interdigitated electrode	DC current between two working electrodes on the polymer film is used to measure electronic conductance; analysis of current transient resulting from potential step is used to separate ion-diffusion from electron conduction	Uses the geometry of the interdigitated electrode array; measure the electronic and ionic conductance by switching between two measurement modes	Analysis of transients require correction for capacitive charging current although these errors can be minimized by using small potential steps; the assumption of faradaic currents being equal at the two working electrode potentials; must be verified for the specific polymer under test
In-Situ measurement Combination of above methods	Polymer thin film on IDM Allows for electrochemical doping in LIB electrolyte 2-probe measurement to determine electronic conductivity in LIB electrolyte at various potentials; 3-electrode measurement to determine ionic conductivity at any given potential	Electronic conductivities up to 10 S cm^{-1} and ionic conductivities up to $10^{-3} \text{ S cm}^{-1}$; can be measured based on electrode geometry; the difference in orders of magnitude between ionic and electronic conductivities does not affect measurement; can be used with different LIB electrolytes	Upper limit of conductivity measurement is dictated by electrode dimensions; polymer should be insoluble in liquid electrolyte

$$\sigma = \Phi_I \sigma_I + \frac{\Phi_N}{\frac{1-\Phi_I-\Phi_N}{\sigma_P} + \frac{\Phi_I}{\sigma_I} + \frac{\Phi_N}{\sigma_N}} \quad (5)$$

Where σ_N , σ_I , and σ_P represents conductivity of the three ion-conducting phases: inorganic fillers, interface between inorganic fillers and polymer, and polymer containing lithium salt. Φ_N and Φ_I represent the volume of inorganic fillers and interfacial region, respectively [78]. The ion conduction mechanisms in solid electrolytes are still not fully understood, and doing so will be the key to developing better solid electrolytes. A combination of density functional theory (DFT) and molecular dynamics (MD) simulations with in-situ or experimental methods are needed to fully understand the correlation between ionic conductivity and composition, chemical structure, electronic structure, and the microstructure of electrolytes. The final goal is to develop solid electrolytes with ionic conductivity of $10^{-3} - 10^{-2} \text{ S cm}^{-1}$, which is comparable to that of liquid electrolytes [54].

3.2. Diffusion coefficient of species

In an electrolyte, diffusion occurs by the movement of ions through the crystal lattice of the electrolyte material. The diffusion of ions in the electrolyte is important for the ionic conductivity of the material. At the interface between the electrode and the electrolyte, diffusion occurs as ions move from the electrolyte into the electrode and vice versa. This is important for the performance of the battery, as it affects the rate at which charge can be transferred between the electrode and the electrolyte. In the electrode, diffusion occurs by the movement of ions through the electrode material. This is important for the capacity of the battery, as it determines the amount of charge that can be stored in the electrode. In polycrystalline materials, such as some solid electrolytes and electrodes, diffusion can occur more quickly along grain boundaries

than through the bulk of the material. This is because grain boundaries can have a higher concentration of defects, which can act as diffusion pathways for ions.

In SSBs, where the anode is composed of lithium metal, the bulk atomic diffusivity of lithium is often the limiting factor in determining the cycling behavior. Fast bulk diffusivity of lithium atoms is required to maintain a stable interface between the lithium metal electrode and the solid electrolyte under realistic cycling current densities. This is due to the movement of lithium ions into the solid electrolyte during stripping, which creates vacancies at the surface of the electrode, which causes contact loss at the atomic scale if the diffusion of lithium atoms to the interface is not fast enough to refill the vacancies. These vacancies can then form voids which lead to macroscopic contact loss and an increase in the electrode-solid electrolyte interface impedance, which can eventually lead to the cell failure during subsequent cycles [79]. Many studies have reported that the internal short circuits can be caused by the localized electrodeposition arising from the inhomogeneous interfacial nature of the electrode and electrolyte, which is influenced by the intrinsically low lattice-lithium diffusivity in dynamic lithium stripping and deposition processes [80]. It is also important to mention that diffusion of elements and chemical reactions at the cathode/SSE interface also hinder Li-ion transportation, and negatively influence battery performance [81].

Diffusion coefficients of species in solid-state batteries are important parameters that affect the battery's performance. In solid-state batteries, the movement of ions or electrons through the solid-state electrolyte or electrode material is primarily determined by the diffusion coefficient of the species. The diffusion coefficient is a measure of how quickly the species can move through the material [82]. Lithium-ion diffusion is a thermally activated hopping process and is typically described in the form [83]:

Table 4
Comparison in activation energy E_a and diffusion coefficients d [88].

Species	$E_a(\text{exp})$ (eV)	$E_a(\text{cal})$ (eV)	$D(\text{exp})$ ($\text{cm}^2 \text{s}^{-1}$)	$D(\text{cal})$ ($\text{cm}^2 \text{s}^{-1}$)
<i>Anode</i>				
Li	0.52–0.57	0.575	$(5-9) \times 10^{-11}$	$(1-10) \times 10^{-11}$
LiC ₆	0.27–0.55	0.283	$(3.8-7.6) \times 10^{-11}$	9×10^{-12}
LiCl ₂	0.17	0.297	1.46×10^{-10}	9×10^{-12}
Li ₄ Ti ₅ O ₁₂	0.2–0.48		$(1.15-6.8) \times 10^{-11}$	
SEI				
LiF	1.99	2.03		3.5×10^{-16}
LiCl	1.47			
LiBr	1.29			
LiI	1.05			
Li ₂ O	0.86			1.6×10^{-16}
Li ₂ S	0.90			
Li ₃ N	0.41–0.43 (ab) 0.68 (c)		2.53×10^{-13} (ab), 3.49×10^{-10} (c)	
Li ₂ CO ₃		0.67		1.1×10^{-7}
<i>Sulfide electrolyte</i>				
Li _{3,25} Ge _{0,25} P _{0,75} S ₄	0.21			
Li _{3,4} Si _{0,4} P _{0,6} S ₄	0.5–0.6			
Li ₇ P ₃ S ₁₁	0.18	0.17		
β -Li ₃ PS ₄	0.16	0.27		
γ -Li ₃ PS ₄	0.22			
Amorphous Li ₃ PS ₄	0.35			
Li ₁₀ GeP ₂ S ₁₂	0.21–0.24	0.22–0.23	2.20×10^{-8}	1.00×10^{-7}
Li ₇ GePS ₈	0.23		1.80×10^{-8}	
Li ₉ P ₃ S ₁₂		0.26		
Li ₁₁ AlP ₂ S ₁₂		0.18		
Li ₁₁ Si ₂ PS ₁₂	0.19		3.50×10^{-8}	
Li ₁₀ SnP ₂ S ₁₂	0.23–0.27	0.24	$1.4-1.8 \times 10^{-8}$	4.40×10^{-8}
Li ₁₀ SiP ₂ S ₁₂		0.2		1.90×10^{-7}
Li ₁₀ GeP ₂ Se ₁₂		0.19		
Li _{9,54} Si _{1,74} P _{1,44} S _{11,7} Cl _{0,3}	0.24			
Li ₆ PS ₅ Cl	0.33			
Li ₆ PS ₅ Br	0.3–0.4			
Li ₆ PS ₅ I	0.3–0.4			
<i>Oxide electrolyte</i>				
Li ₁₀ GeP ₂ O ₁₂		0.36		
Li ₇ La ₃ Zr ₂ O ₁₂	0.32	0.25		
Li ₅ La ₃ Bi ₂ O ₁₂	0.47			
Li ₆ SrLa ₂ Bi ₂ O ₁₂	0.42			
Li ₅ La ₃ Nb ₂ O ₁₂	0.43			
Li ₅ La ₃ Ta ₂ O ₁₂	0.56			
Li ₆ SrLa ₂ Ta ₂ O ₁₂	0.50			
Li ₆ BaLa ₂ Ta ₂ O ₁₂	0.40–0.44			
Li _{6,4} La ₃ Zr _{1,4} Ta _{0,6} O ₁₂	0.35			
Li ₆ La ₃ Ta _{1,5} Y _{0,5} O ₁₂	0.36			
Li ₆ CaLa ₂ Ta ₂ O ₁₂	0.50			
Li ₆ La ₃ HfNbO ₁₂				
Li ₆ La ₃ SnSbO ₁₂	0.50			
Li ₆ La ₃ SnNbO ₁₂	0.50			
Li ₆ La ₃ SnTaO ₁₂	0.50			
Li ₆ La ₃ ZrTaO ₁₂	0.42			
Li ₆ La ₃ ZrNbO ₁₂	0.40			
Li ₇ La ₃ Hf ₂ O ₁₂	0.29			
Al-doped Li ₇ La ₃ Zr ₂ O ₁₂				
Li _{6,25} Ga _{0,25} La ₃ Zr ₂ O ₁₂	0.25			
Li _{6,25} La ₃ Nb _{1,375} Sc _{0,625} O ₁₂	0.36			
Li _{6,5} La ₃ Zr _{1,75} Te _{0,25} O ₁₂	0.37			
Li _{7,06} La ₃ Y _{0,06} Zr _{1,94} O ₁₂	0.26			
Li _{1,3} Ti _{1,7} Al _{0,3} (PO ₄) ₃		0.25		
Li _{1,2} Ti _{1,8} Al _{0,2} (PO ₄) ₃	0.22–0.28			
Li _{2,88} PO _{3,73} N _{0,14}	0.45–0.55			
LiHf ₂ (PO ₄) ₃	0.33			
LiTi ₂ (PO ₄) ₃	0.48			
LiGe ₂ (PO ₄) ₃	0.60			
Li _{3,5} Si _{0,5} P _{0,5} O ₄	0.49			
Li _{1,4} ZnGe ₄ O ₁₆	0.5	0.32		
Li _{0,31} La _{0,56} TiO ₃	0.33	0.26		
<i>Halide electrolyte</i>				
Li ₃ InCl ₆	0.326–0.347	0.20		
Li ₃ ErCl ₆	0.41			
Li ₃ YCl ₆	0.40–0.45	0.19		
Li ₃ YBr ₆	0.37	0.28		
LiScCl ₄	0.39	0.37		
Li _{1,5} ScCl _{4,5}	0.38			

(continued on next page)

Table 4 (continued)

Species	$E_a(\text{exp})$ (eV)	$E_a(\text{cal})$ (eV)	$D(\text{exp})$ ($\text{cm}^2 \text{s}^{-1}$)	$D(\text{cal})$ ($\text{cm}^2 \text{s}^{-1}$)
$\text{Li}_{1.8}\text{ScCl}_{4.8}$		0.34		
Li_2ScCl_5	0.38			
$\text{Li}_{2.5}\text{ScCl}_{5.5}$	0.37			
Li_3ScCl_6	0.36	0.18–0.25		
$\text{Li}_{3.5}\text{ScCl}_{6.5}$	0.36			
Li_4ScCl_7	0.38			
Li_5ScCl_8		0.26		
Li_3ScBr_6		0.30		
Li_3HoCl_6		0.19		
Li_3HoBr_6		0.26		
$\text{Li}_{2.5}\text{In}_{0.5}\text{Zr}_{0.5}\text{Cl}_6$		0.20		
$\text{Li}_{2.5}\text{Sc}_{0.5}\text{Zr}_{0.5}\text{Cl}_6$		0.16		
LiAlCl_4	0.47–0.50			
LiAlBr_4	0.437			
LiAlI_4	0.429			
$\text{Li}_2\text{Sc}_{2/3}\text{Cl}_4$		0.277		
$\text{Li}_2\text{Sc}_{2/3}\text{Br}_4$		0.366		
$\text{Li}_2\text{Sc}_{2/3}\text{I}_4$		0.257		
Cathode				
LiCoO_2	0.3	0.31	10^{-16} – 10^{-11}	
LiNiO_2	0.59		7.3×10^{-19}	
$\text{Li}_{1.2}\text{Co}_{0.13}\text{Ni}_{0.13}\text{Mn}_{0.54}\text{O}_2$	0.53			
$\text{Li}_{1.2}\text{Ni}_{0.15}\text{Co}_{0.1}\text{Mn}_{0.55}\text{O}_2$	0.26			

Table 5

Material level thermal characteristics for solid electrolyte classes.

	Decomposition Temperature	Decomposition Products	Possible Improvements
Polymer	200–600°C, dependent on material composition, crystallinity, testing atmosphere [96].	Decomposition products vary based on the balance of gas components in the testing atmosphere [104].	Addition of PTEE (polytetrafluoroethylene) reduces flammability of PEO (polyethylene oxide) [105]. Composite coordination (copolymerization, incorporation of inorganic materials).
Oxide	800–1600°C due to valence electron distributions and availability of energetically stable configurations [97].	Possible formation of AlPO_4 impurities or carbon isotopes [106]. Risk for formation of reactive byproducts.	Reduction of AlPO_4 byproduct attributed degradation via Nb doping to prevent structural deformations [106].
Sulfide	700–1700°C due to the formation of more energetically stable structural phases [99].	Formation of hazardous (flammable, combustible, etc.) byproducts, including highly toxic H_2S gas [98].	Si-substituted $\text{Li}_{3.25}\text{Si}_{0.25}\text{P}_{0.75}\text{S}_4$ can be implemented to maintain β structure up to 600°C [107]. Doping, such as the addition of Sn into Li-Sn-S systems can increase thermal stability [108].
Halide/Hydride	300–400°C, dependent on atmospheric conditions and material interactions [109].	Hydride solid electrolytes can decompose to generate H_2 byproducts [109].	Doping could be performed to improve conductivity, stability, or safety. Stabilization approaches require improvement to balance safety and performance [103].

$$D_1 = D_0 e^{\frac{-E_A}{RT}} \quad (6)$$

Where D_0 is the exponential prefactor, E_A is the activation energy of diffusion, T is the temperature in Kelvin, and k_B is the Boltzmann constant. The equation fundamentally assumes a well-defined activation energy for the hopping of atoms or ions between specific lattice sites in a crystalline material. While amorphous phases like glassy Li_3PS_4 lack long-range order and distinct lattice sites, this model can still provide a reasonable approximation of the temperature dependence of ionic conductivity (which is directly related to the diffusion coefficient via the Nernst-Einstein relation) [84]. In amorphous solids, diffusion occurs through a distribution of energy barriers and jump distances within the disordered network structure. The experimentally observed diffusion behavior in glassy Li_3PS_4 often exhibits Arrhenius-like behavior over certain temperature ranges, suggesting that a dominant average activation energy governs the transport process [85]. However, deviations from ideal Arrhenius behavior, such as curvature in the Arrhenius plot, can arise due to the distribution of energy barriers and potential changes in the diffusion mechanism with temperature in the disordered structure. Therefore, while the Arrhenius equation offers a useful simplified description, a more complete understanding of diffusion in amorphous materials requires considering the inherent structural disorder and the resulting distribution of energy landscapes. To determine the diffusion coefficient of charge-carrying ions in a solid electrolyte, a structure is fabricated and then studied using impedance spectroscopy [86]. Analysis of the impedance of the assembled structure leads to determining

the Warburg coefficient and amplitude. The Warburg amplitude value is related to the diffusion coefficient of the charge-carriers by the following equation:

$$A_{w1} = \frac{U_0}{\sqrt{2D_1 S q c_0}} \quad (7)$$

Where U_0 is the voltage amplitude across the electrodes, D is the diffusivity coefficient, S is surface area, q is ion charge, and c_0 is the amplitude of oscillation of the non-equilibrium concentration of charge-carrying ions near the electrode. Solving for the diffusion coefficient in this equation, and replacing c_0 , results in the following equation:

$$D_1 = \frac{1}{2} \left(\frac{d\delta}{A_{w1} 2S\epsilon_0\epsilon_r} \right)^2 \quad (8)$$

Where d is the thickness of the electrolyte layer and ϵ_r is its relative dielectric permeability, and δ is the distance to the electrode. There are multiple methods that can be used to determine the diffusion coefficient of charge-carrying ions in electrode materials. The galvanostatic intermittent titration technique (GITT) is the most widely used method. However, GITT can be time-consuming and may require sophisticated interpretation models. GITT consists of two repeating steps: (1) a constant current is applied for a duration where the assumption of semi-infinite diffusion holds, (2) the current is switched off until the voltage becomes invariant. The analysis of the electrode potential that was measured during the current pulse and the change in the equilibrium potential results in the chemical diffusion coefficient of the charge-

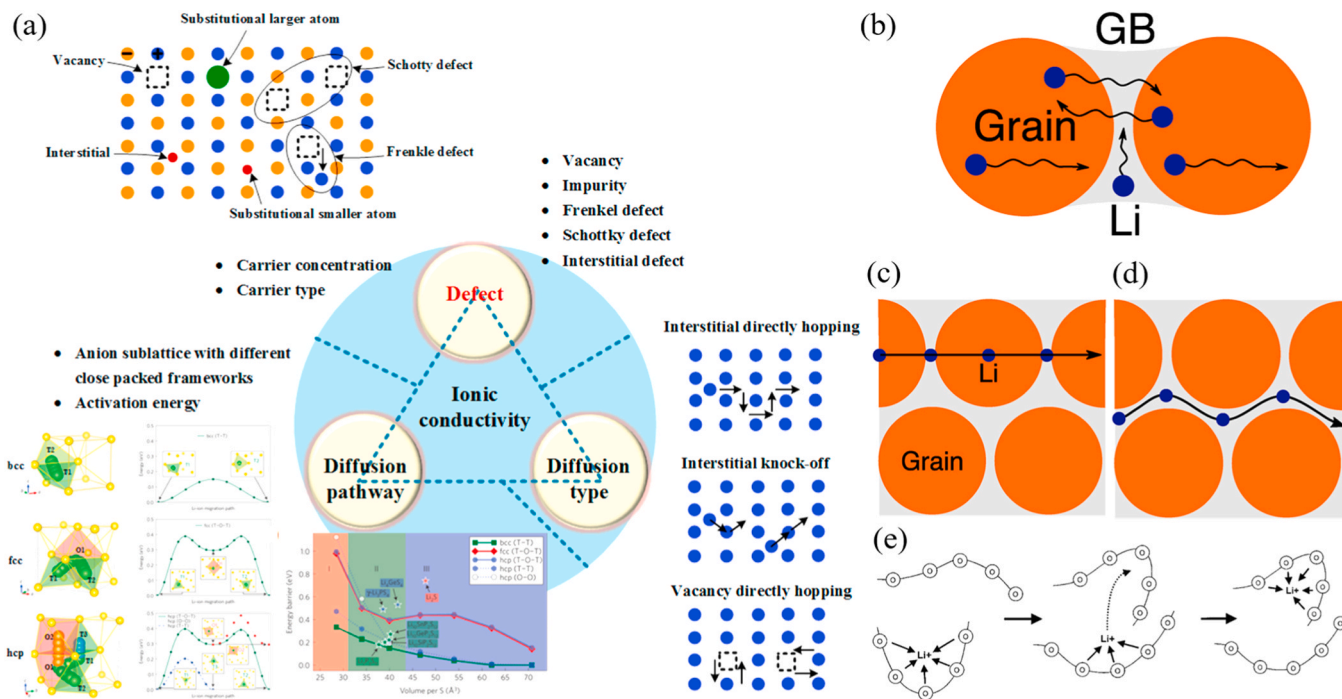


Fig. 4. (a) The defect, migration pathway, and migration mechanisms behind ionic conductivity in ceramic solid electrolytes [72]. possible transport mechanisms that occur at (b) grain boundaries, (c) granular conduction, and (d) grain boundary conduction [73]. (e) Li^+ transport mechanism found in polymers, which is defined as segment motion [74].

carrying ions. The intermittent current interruption (IC) method is another method that can be used. For IC, repeating transient current interruptions are applied to the cell while it is under constant-current cycling. Then, through linear regressions of the potential change against the square root of step time during the current pauses, the time-independent and time-dependent parts of the resistance are calculated, which are also called internal resistance and diffusion resistance coefficient. This can also characterize the diffusion processes. Both methods calculate the diffusion coefficient, D , using the following equation:

$$D = \frac{4}{\pi} \left(\frac{V \frac{\Delta E_{OC}}{\Delta t_i}}{A \frac{dE}{d\sqrt{t}}} \right)^2 \quad (9)$$

Where V is the molar volume of the electrode materials, A is the surface area of the electrode, E_{OC} is the open circuit potential, Δt_i is the period when a constant current is applied between consecutive open circuit potential measurements, E is the electrode potential, and t is the step time [87]. Molecular dynamics simulation is also a tool that can be used to study transport phenomena at the molecular level.

There are possible methods being used and currently being developed to increase the diffusion of species through SSBs. Reducing particle size of the solid-state electrode and the electrolyte can increase surface area and decrease the diffusion distance. Increasing the porosity of the electrode can increase surface area available for ion transport. Adding conductive additives to improve electrode conductivity which can decrease polarization and improve the diffusion coefficients. Introducing defects (vacancies or interstitials) can create additional diffusion pathways for ions. However, too many defects can decrease stability and durability of the electrolyte. Controlling the diffusion of species heavily affects the ionic conductivity of an SSE. Being about to design materials with specific diffusion rates could be the answer to developing materials with higher ionic conductivity. Increasing diffusion rates of lithium ions will have a big impact on decreasing dendrite formation [89]. Fully understanding diffusion mechanisms and being able to control diffusion rates will ultimately lead to better performing SSBs.

3.3. Thermal issues in SSBs

Thermal issues are a key concern when considering the practical performance and overall safety of batteries and battery-integrated systems. Though not limited to solid state batteries alone, thermal issues at the material, interface, and battery level present complex interactions of electro-chemical factors that contribute to decision making regarding battery manufacturing approaches and impact device functionality. Solid state batteries are, overall, understood to improve upon the thermal stability of traditional liquid electrolyte systems regarding volatility, flammability, and flowability, however the testing of solid-state thermal interactions remains within its early stages, and development must continue to allow for a full understanding of their general thermal behavior.

A primary concern for the thermal performance of solid-state batteries is the risk of thermal runaway: a chain reaction of thermal interactions that results in an uncontrolled temperature increase. Thermal runaway can occur when the battery temperature exceeds the appropriate working temperature, causing the release of reactive byproducts and, at best, leading to detriment of battery performance via damage or melting, or, at worst, causing full device combustion or explosion that could lead to user injury and even loss of life. In 2019, in a thermal runaway event, caused by the failure of a single lithium-ion battery cell due to “abnormal lithium deposits”, resulted in a large fire, explosion, and the injury of four firefighters [90]. This incident helped bring attention to the need for improved battery storage considerations for thermal runaway prevention measures such as barriers to propagation and a standardized response procedure. In the design of solid-state batteries thermo-chemical interactions must be analyzed carefully and comprehensively to allow a complete understanding of how the battery system behaves under a range of thermal conditions at all levels.

In addition to thermal runaway, researchers, developers, and manufacturers of solid-state batteries must also consider the possibility, and subsequent consequences, of morphological changes at the electrolyte level caused by thermal exposure. Melting, shrinkage, and cracking of solid electrolyte components under heating conditions can hinder

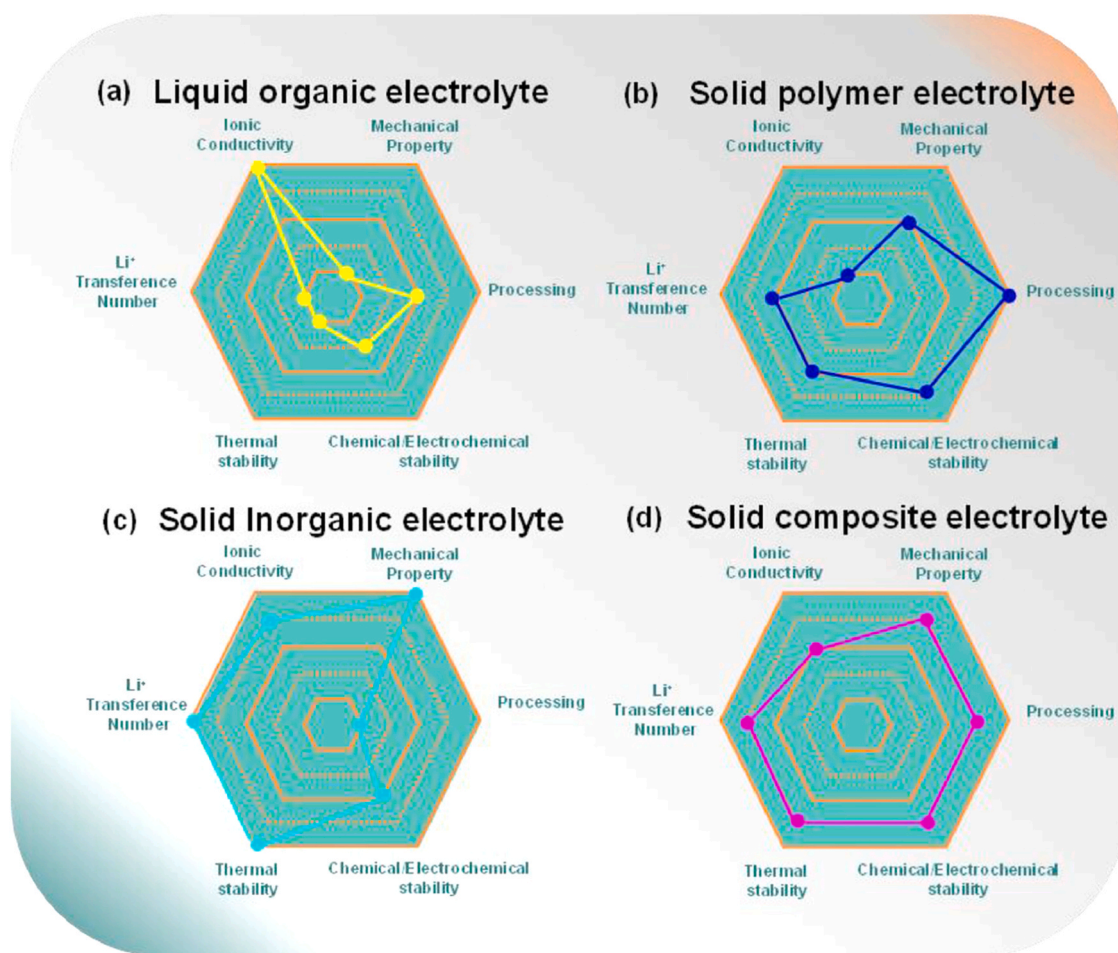


Fig. 5. Composite solid electrolytes offer many improvements over other types of solid electrolytes [77].

electrochemical performance and ionic conductivity by altering the interfacial interactions between the solid electrolyte and electrodes within the battery. Additionally, the application of thermal variability to solid-state batteries can cause material level decomposition behaviors that could result in the emission of hazardous byproducts, leading back to the possibility of thermal runaway and potential user harm [91]. Careful consideration of thermal safety at all levels for solid state battery design, from material selection to interface and battery level interactions, allows for the design and manufacture of solid-state systems that permit for the maximization of energetic performance and user safety.

3.3.1. Materials level analysis of thermal stability

Within the field of solid-state battery development there exist several key categories that are defined by their primary solid electrolyte material selection: polymer, oxide, sulfide, hydride, and sulfide. Each possible material selection provides a unique set of thermal characteristics and interactions that define their feasibility and performance within overall battery function. Primary temperature testing must provide definition for the key transition temperatures of the electrolyte material alone. These values include crystallization temperature (T_C), glass transition temperature (T_G), melting point (T_M), and decomposition temperature (T_{dec}), which can be calculated using Differential Scanning Calorimetry (DSC) or Differential Thermal Analysis (DTA). These analysis methods utilize comparison of thermal behavior of a sample with a known reference material to determine the temperatures at which exothermic and endothermic events occur, indicating occurrence of reaction or transition events. As indicated in Fig. 6b the overall decomposition temperatures of solid-state electrolytes are, for the most

part, higher than those of liquid batteries, with oxide and sulfide electrolytes offering the highest overall decomposition temperatures. Thermal stability characteristics are mostly determined by structural properties of the material such as crystallinity, elemental combination, and even degree of polymerization in the case of polymers. Material level thermal stability can become a concern when thermal exposure results in the morphological alteration of the electrolyte material, or the release of hazardous or reactive byproducts.

Polymer materials such as polyethylene oxide (PEO) and polymethyl methacrylate (PMMA) are common choices in solid state polymer electrolyte design for their high interfacial contact capabilities and chemical stability, however, polymer materials suffer from relatively low temperature thermal decomposition, typically below 400°C [96]. Specific polymer electrolyte decomposition and melting patterns are largely influenced by the material composition, polymer chain length, and polymer chemical structure. In some cases, composite polymer electrolytes are designed to integrate coordination between different polymer types, or between polymers and inorganic materials, to improve electrochemical performance or thermal stability. Magnesium aluminate nanoparticle fillers, for example, have been shown to raise the decomposition temperature of polyethylene oxide lithium hexafluorophosphate (PEO-LiPF₆) from about 200°C to above 300°C. In addition to selective design of composite polymer electrolytes, thermal stability at the material level can be further improved through careful selection of fabrication methods as a means of controlling the polymer crystallinity as it is intimately related to thermal decomposition temperature. Construction of multilayer solid electrolytes can inhibit interfacial reactivity while improving thermal stability, and heat treatment processing can stabilize thermal reactivity by decreasing

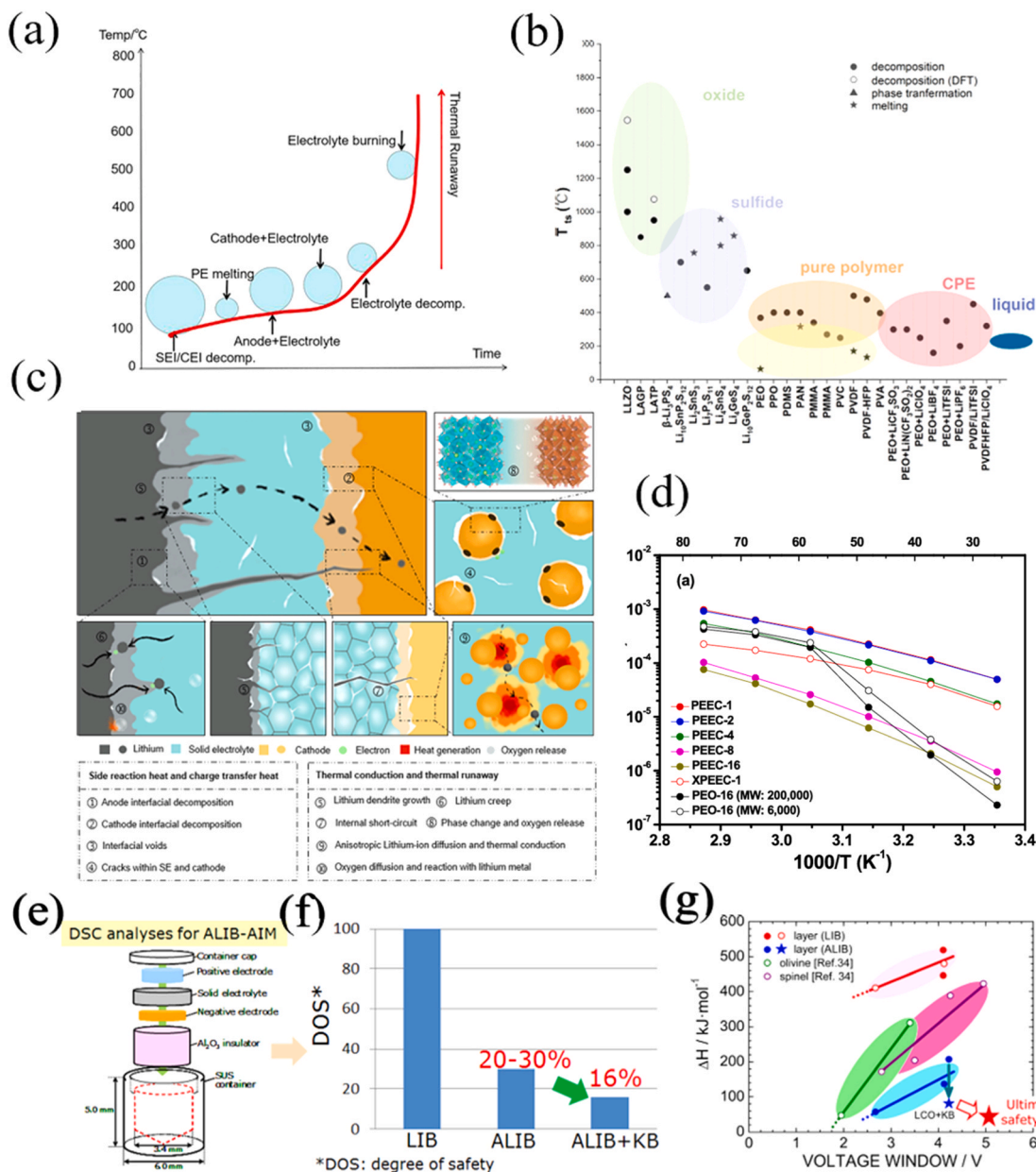


Fig. 6. (a) A qualitative representation of interactions that can lead to thermal runaway in solid state battery systems [92]. (b) a diagram comparing key decomposition factors for a variety of solid electrolytes of different material types [91]. (c) summary of thermally related interfacial interactions and possible subsequent complications [93]. (d) temperature dependence of ionic conductivities of various solid polymer electrolytes related interfacial compositions [94]. (e) schematic showing the design of all solid-state lithium-ion battery (ALIB) all-inclusive-microcell (AIM) used to examine overall battery safety under variable thermal conditions [95]. (f) a bar chart comparing the degree of safety (DOS) of liquid LIB, all solid-state lithium-ion batteries (ALIB), and ALIB with the addition of Ketjenblack (KB) and oxygen stabilizer. Ultimate safety occurs when DOS is at 0% [95]. (g) a "safety map" visualizing changes in heat generation (ΔH) based on operating voltage (voltage window) from differential scanning calorimetry (DSC) tests [95].

crystalline impedance [96].

Among the material categorizations of solid electrolytes, oxides provide the highest level of thermal stability with decomposition temperatures ranging from 800 to 1600°C due to their valent electron distributions and whether energetically stable configurations can be formed [97]. The decomposition temperature for LATP(Li_{1.5}A_{10.5}Ti_{1.5}(PO₄)₃), for example, is as high as 1394 K or about 1120°C. Despite their high thermal stability, oxide material electrolytes suffer from morphological deformation with increasing temperature exposure. These electrolytes undergo transitions in response to heat treatment that cause structural defects such as shrinking and cracking that can interfere with interfacial stability and efficiency of ionic conductivity. In the case of ATP(Li_{1.5}A_{10.5}Ti_{1.5}(PO₄)₃) morphological deformation is primarily

the result of the AlPO₄ transitional production, and researchers have had success in minimizing this production using Nb (niobium pentachloride) doping [33]. Overall, although oxide solid electrolytes can provide a high standard for thermal stability, formation of reactive byproducts and risk of morphological deformation must be accounted for while electrochemical and thermal properties are optimized.

Similar to oxide solid electrolytes, sulfide solid electrolytes provide high degradation temperatures, however, their thermal decomposition may result in the formation of unfavorable byproducts, the most concerning being highly toxic H₂S gas [98]. When heated, sulfide electrolytes typically experience phase transitions that can alter ionic conductivity. Li₃PS₄, for example, has three structural phases: α , β , and γ with the β phase converting to α at about 500°C, and no longer being

capable of reverting to β , but instead forming the γ structure during the cooling process, with an ionic conductivity of only $3 \times 10^{-7} \text{ S cm}^{-1}$ [99]. Structural alteration upon heating has also been observed, such as in the case of $\text{Li}_7\text{P}_3\text{S}_{11}$, which loses its structure after heating to 550°C to be replaced by phases that offer higher thermal stability [100]. Doping of sulfide solid electrolytes, such as using Sn or lithium salts, has been proposed as a method to increase glass stability temperature, however, doping must be done selectively as improper material selection can also shift the temperature and crystallinity behaviors unfavorably. Additionally, electrolyte manufacturing must be considered as the preparation method used can impact the glass stability without significantly altering the ratio between the glass stability and crystallization value [91].

Finally, halide and hydride solid electrolytes provide a unique set of material characteristics. Although these electrolytes offer promising ability to retain structure and increase ionic conductivity upon thermal heating (up to nearly 400°C), most heating beyond 300°C results in disorganization of crystallinity [55]. Some halide solid electrolytes also exhibit unfavorable structural transformations at relatively low temperatures at which they are susceptible to decomposition and fail to recrystallize with sound structure [101]. Hydride solid electrolytes can also behave unfavorably, such as LiBH_4 which suffers from poor thermal stability, causing detriment to ionic conductivity at only 115°C [102]. Although some halide and hydride solid electrolytes have promise for solid state battery development, they are, at present, unfavorable due to relatively low temperature instability, risk of dehydrogenation, and risk for production of flammable gas byproducts. Stabilization of thermal degradation byproducts from halide/hydride solid electrolytes is required to develop solid-state batteries that are safe and effective with electrolyte selection [103].

3.3.2. Interface level analysis of thermal stability

Beyond solid electrolyte material selection, interfacial interactions between the electrolyte and its electrode components (anode and cathode) introduce an additional subset of thermal interactions with implications for performance capabilities and overall safety. As previously described, thermal treatment of electrolyte materials can cause alterations in material morphology, stability, and crystallinity, all of which can impact interfacial interactions. Fig. 6c visualizes several prominent interfacial interactions and alterations that occur under variable thermal conditions, alongside the possible consequences such as thermal conduction interference and thermal runaway. Similar to material level thermal interactions, interface level interactions can be analyzed using differential thermal analyses to observe transition events among electrolytes with different electrode component selections applied.

Testing of polymer solid electrolytes with lithium metal anodes and a variety of appropriate cathodes indicated that no strong interactions between the polymer and anode occurred to cause material deformation, however, contact with the cathode acceleration of material decomposition attributed to the open-circuit voltage of the system [110]. Fig. 6d gives the temperature dependence of ionic conductivities of various solid polymer electrolytes with different compositions [94]. In addition to detriment to ionic conductivity, thermal influence on interfacial interactions can also introduce thermal runaway behavior such as in the case of $\text{Li}_6\text{PS}_5\text{Cl}$ in $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ SPE battery systems. This detrimental reactivity occurs due to chemical reactivity between delithiated $\text{Li}_{1-x}\text{Ni}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ and the solid electrolyte. This violent reaction is, however, absent in systems using LiFePO_4 as the electrolyte is stable with delithiated $\text{Li}_{1-x}\text{FePO}_4$ showing that issues with thermal runaway can be circumvented through careful design and selection of electrolytes alongside electrode components [111].

Generally, oxide solid electrolytes suffer from insufficient interfacial contact as well as high impedance levels and interface-level thermal stability in this material classification is largely dependent on the constituent elements and structures of the solid electrolytes being used. Layered cathode structures have been observed to decompose into

transition metal oxides that can form byproducts with increasing thermal conditions and some oxide solid electrolytes were even reported to react violently with phosphate electrodes under high temperature conditions [112]. Specific interphase compositions like Li_2CO_3 and Li_3PO_4 at the cathode/SSE interface are identified using a combination of surface-sensitive analytical techniques. X-ray Photoelectron Spectroscopy (XPS) is a crucial method, as it can determine the elemental composition and chemical states of the interfacial layers by analyzing the core-level electron binding energies. The presence of Li_2CO_3 is typically indicated by characteristic C 1 s and O 1 s peaks corresponding to carbonate species, while Li_3PO_4 is identified through P 2p and O 1 s peaks associated with phosphate [113,114]. Complementary techniques such as Secondary Ion Mass Spectrometry (SIMS) provide information on the elemental distribution and depth profiling across the interface. Furthermore, in-situ and operando techniques, like Electrochemical Impedance Spectroscopy (EIS) coupled with mass spectrometry (DEMS), can correlate electrochemical changes with the evolution of gaseous products (e.g., CO_2 from carbonate decomposition) and interfacial resistance changes, indirectly confirming the presence and degradation of these phases during battery cycling [115]. Thermal runaway among oxide electrolyte materials was first observed in LAGP-Li whereas melting of LAGP in argon gas resulted in “violent combustion” from the release of O_2 . Thermal runaway risks have also been observed in LLTO and the thermal runaway pattern may be the result of O_2 production causing exothermic reactivity and melting of lithium materials [116]. Interfacial thermal stability of oxide solid electrolytes could possibly be improved by elemental doping treatments; however, such a possibility requires further testing for confirmation.

Little work has been done to examine the interfacial thermal stability patterns of sulfide solid electrolytes, likely due to the risk of toxic gas production (mainly H_2S). Based on the available thermal studies on sulfide solid electrolytes, interfacial thermal stability is understood to be influenced by cathode state of charge, atmospheric pressure conditions, and charge/discharge cycle number. Some researchers propose that suppressing the exchange of sulfur and oxygen and selecting oxygen stable sulfide solid electrolytes, could increase interfacial thermal stability [100]. To understand more about the thermal interactions of sulfide solid electrolytes a relatively recent research effort proposed a new method to examine the thermal stability pathways of sulfide solid-state electrolytes using high temperature sulfur precipitation [117]. The study proposes a new conceptual parameter ‘Th’, a term to account for the different chemical bond characteristics contained in sulfide solid electrolytes. Although this analytical strategy for the analysis of sulfide electrolyte stability has not yet been widely adopted, it presents an opportunity to better understand thermal patterns to better guide sulfide SE production and verification strategies [117].

Although halide solid electrolytes, as previously described, have high thermal stability, their interfacial thermal stability is complicated by cathode and anode material selections with complex metal ion interactions. Halide interactions without metal ion interactions, however, such as a $\text{Li}/\text{Li}_2\text{OHCl}/\text{Li}$ cell, have shown promise for providing good stability at high working temperatures [118]. Thermal runaway risk trends require further analysis in the case of halide/hydride solid electrolytes as the majority of test results are circumstantial, encompassing only a select few specific materials rather than the material class as a whole. For instance, a 2020 study of $\text{Li}(\text{CB}_9\text{H}_{10}) - \text{Li}(\text{CB}_{11}\text{H}_{12})$ complex hydride solid electrolytes indicated both endothermic and exothermic peaks during heating-cooling cycles, however no largely detrimental runaway patterns were reported [119]. Further work must be performed to examine the possible risks of thermal runaway in halide and hydride solid electrolytes prior to commercial development in solid-state battery systems.

Composite electrolytes are designed to enhance mechanical properties, but the mismatch in thermal expansion coefficients (CTEs) between polymer and ceramic phases in composite materials significantly impacts interfacial stability under thermal cycling, especially at elevated

temperatures [120]. Polymers typically exhibit much higher CTEs compared to ceramics due to their weaker intermolecular bonds. During heating, the polymer phase will expand to a greater extent than the ceramic phase, leading to the development of stresses at the interface. Upon cooling, this difference in contraction rates induces further stress. Repeated thermal cycling exacerbates these stresses, potentially causing interfacial debonding, cracking, and ultimately, the failure of the composite. At elevated temperatures, the mechanical properties of the polymer matrix often degrade, reducing its ability to withstand these stresses and further compromising the interfacial integrity. The magnitude of the stress is directly proportional to the difference in CTEs and the temperature change, highlighting the critical role of CTE matching for long-term stability [120].

3.3.3. Battery level analysis of thermal stability

Among all levels of thermal analysis for solid state batteries, whole battery performance testing remains the least developed, despite being heavily important. Battery level testing for liquid electrolyte systems has typically consisted of whole battery thermal runaway analysis using differential scanning calorimetry (DSC) and/or accelerating rate calorimetry (ARC) to characterize heat release behaviors among all battery components (electrolyte, cathode, anode, and separator).

In a study about battery-level thermal performance capabilities, researchers performed comparative analysis on Li/LiFePO₄/I whole-battery systems to compare thermal performance of a polymer solid electrolyte with a traditional liquid electrolyte [91]. This experiment showed that the liquid electrolyte battery generated twice the amount of heat than that of the polymer LIB, and the polymer LIB was also indicated to provide overall higher thermal stability. Other studies have corroborated that, solid electrolytes can provide lower self-heating rates than liquid electrolyte batteries while others report mixed results. For example, a study of Li_{6.75}La₃Zr_{1.75}Nb_{0.25}O₁₂ using an all-inclusive microcell testing method demonstrated that, for electrolyte-electrode behavior alone, high thermal stability was observed (except at approximately 185°C when lithium fusion in LLZNO occurred) [95]. Despite the initially promising thermal stability, testing of the whole battery system indicated unexpected exothermic peaking due to oxygen release. Based on the battery-level observations collected thus far, further work must be done to examine overall thermal performance of solid-state batteries and to ensure that whole battery testing adequately encapsulates overall safety performance.

3.4. Mechano-chemical analysis

Solid state electrolytes for battery development provide a promising alternative to conventional liquid electrolyte batteries by providing improved energy storage capabilities. The interfacial relationships within solid electrolyte systems, however, introduce a unique set of mechano-chemical interactions that must be thoroughly analyzed and understood for the design of battery systems that maximize efficiency of ionic conductivity while minimizing the risks of dendrite formation, morphological alteration, and undesirable coupling effects.

Dendrites are branch-like projections of lithium metal that form on the surface of the solid-state battery anode and can infiltrate the electrolyte material. Dendrite formation is dependent on mechano-chemical interactions of the anode and electrolyte, and can be influenced by temperature, interfacial elements, and bulk microstructural features [130]. Dendrite growth can lead to short circuiting of solid-state batteries under high current density, despite the fact that solid-state batteries are often praised for their high energy density, and interfacial resistance continues to increase due to solid-solid contact interface when compared to liquid lithium battery systems [41]. Understanding of dendrite growth mechanics via mechano-chemical analyses can allow for control of dendrite growth events in solid-state electrolytes.

In addition to dendrite formation, other types of morphological alteration can occur as the result of mechano-chemical interactions in

solid-state electrolytes. Morphological stability is best maintained when there is a balance between the electrochemical and mechanical properties of the solid electrolyte. More specifically, these results were shown in a study that looked at the interaction between a metal electrode and binary solid polymer electrolyte through mechano-chemical analysis at steady state [126]. The researchers found that there are many potential factors that can influence the morphological stability of a system and that simply increasing the mechanical stiffness of a material does not mean better results. This is because an increase in mechanical stiffness alone may inevitably lead to a decrease in transport properties, which is extremely undesirable as it can then lead to unwanted morphological alterations, such as protrusion branching (which takes place at the perimeter of a protrusion during an extremely elevated current density value). Some other factors that have the potential to influence morphological stability include the coupling effects of “the normal stress dependency of the equilibrium potential, the mechanical influence onto the electrochemical potential in the electrolyte bulk as well as the value of the exchange current density” [126]. Overall, the study showed that both the charging rates and local modifications of the interface conductivities play a role in altering the morphological stability plots between the lithium and polymer interfaces.

3.4.1. How mechano-chemical analysis is performed

Mechano-chemical properties and interactions can be complex to define, therefore, a variety of technical and mathematical methods are utilized to characterize the nature and effects of chemical interactions on the mechanical properties of solid electrolytes. The elastic modulus mismatch between SSEs and electrode materials is a critical factor influencing the mechanical stability and performance of all-solid-state batteries. This mismatch can lead to interfacial stresses, delamination, and capacity fade during battery cycling due to volume changes in the electrode materials. Researchers have employed various experimental and computational techniques to check or quantify this mismatch [131]. Experimentally, techniques like nanoindentation, ultrasonic sound velocity measurements, and uniaxial testing are used to determine the elastic moduli of individual components. Furthermore, in-situ and operando methods such as X-ray diffraction (XRD) and electrochemical strain microscopy (ESM) can provide insights into the stress and strain evolution at the interfaces during battery operation. Computationally, Density Functional Theory (DFT) and Finite Element Method (FEM) simulations are utilized to model the mechanical behavior of the interfaces and quantify the stress distribution arising from the elastic modulus mismatch [132]. These simulations often incorporate experimentally determined elastic moduli and thermal expansion coefficients to provide a more accurate representation of the system. Based on known interactions between lithium dendrite formation and grain boundaries, analytical modeling methods can be used to understand the specific nature of dendrite growth to allow guidance for methods to reduce dendrite formation. This approach, described in a 2017 study by R. Raj and J. Wolfenstine [127] uses Gibbs free energy calculations to produce limit diagrams with “safe” and “fail” regimes to analyze dendrite nucleation patterns. More specifically, Rishi Raj and Jefferey Wolfenstine both introduce the use of current limit diagrams in determining the dendrite formation [127]. Solid-state electrolytes offer appealing quality factors over organic liquid electrolytes such as non-flammability and ability to remain stable with Li-metal. However, solid electrolytes must have high ionic, low electronic conductivity, and a high shear modulus as desirable features to oppose dendrite formation [127]. The purpose of the current limit diagrams is to analyze the dendrite nucleation of the solid-state electrolyte - Li interface [127]. These charts will provide insight into the nucleation of lithium dendrites at the lithium anode-solid state electrolyte interface. The diagram will indicate a “safe” and “fail” regime for battery function. The safe regime suggests the chances of dendrite formation are unlikely whereas the “fail” regime says otherwise. The governing equation used for the limit diagrams is initially derived from the difference in the Gibbs Free Energy

of two states: the supersaturated state and the state where the supersaturation is relieved by the growth of a precipitate, both shown in the Fig. 7c [127]. The governing equation for the limit diagram is the calculated result of the critical current density result (j^*) [127]:

$$j^* = \frac{(2\gamma\Omega_{Li}/r_c) + \frac{\sigma_F\Omega_{Li}}{f} \frac{1}{d(1-\alpha)|e|}}{\rho} \quad (10)$$

Where j^* is critical current density, $|e|$ is positive charge on Li-ion, γ is specific energy of the metal-electrolyte interface, Ω_{Li} is the number of Li atoms per unit volume has a relation with the volume per lithium atom in the electrolyte, σ_F is mechanical resistance to fracture, $1/\rho$ is ionic conductivity of electrolyte, f is the measure of the contribution of grain matrix resistivity to total resistivity, and α is proximity factor measuring how close a grain boundary is to the Li metal interface, as a fraction of grain size. This governing equation defines the transition from a “safe” to a “fail” regime. In Fig. 7d, the following were considered: (1) The ranging factors were ionic conductivity of electrolyte ($1/\rho$) and mechanical resistance to fracture (σ_F), (2) The proximity factor (α) was taken to be 0 and 0.5, (3) The contribution of grain matrix resistivity to total resistivity parameter (f) was taken to be 1.0 and 0.5. As mentioned previously, the Fig. 7d suggests how the critical values for current density (j^*) vary with the solid-state electrolyte’s ionic conductivity ($1/\rho$). From observation, the current limit diagrams provide the following information: (1) Lower fracture stress increases the probability of dendrite formation [127], (2) The safe region also contracts if the grain boundary is closer to the Li-metal interface (smaller α value) [127], (3)

Higher grain boundary resistance increases the chance of formation (higher f value) [127]. (4) High ionic conductivity of the electrolyte is expected to reduce the chance of dendrite formation [127]. The diagrams are consistent with the initial analysis that higher ionic conductivity and lower grain boundary resistance reduce the chance of dendrite formation [127], which is essential information for the operation of solid-state batteries.

Other analyses have also been performed to capture the trigger mechanisms resulting in the coupling effects of morphological alteration and protrusion growth. More specifically, the same group of researchers further expanded their experiment by using mechano-chemical analysis to now look at the relationship between a solid electrolyte and metal electrode interface during charging and discharging [126]. This is all done with hopes of better understanding the possible reasons for the unwanted development of inhomogeneous material deposition and interface conductivity directly contributes to the damages/ alterations of the solid electrolyte’s morphology. There were four hypotheses outlined for the potential trigger mechanisms that may contribute/ lead to the development of inhomogeneous deposition and shown below.

Hypothesis 1 suggests that a harmful protrusion growth in the separator is caused by an initial movement/ protrusion due to surface roughness and directly affects the transport paths. Hypothesis 2 assumes that the cathode consists of an inhomogeneous current supply and therefore lacks active storage particles. This then leads to the mirroring of the morphology of the positive electrode on the negative electrode and therefore increases the amount of applied current density by two

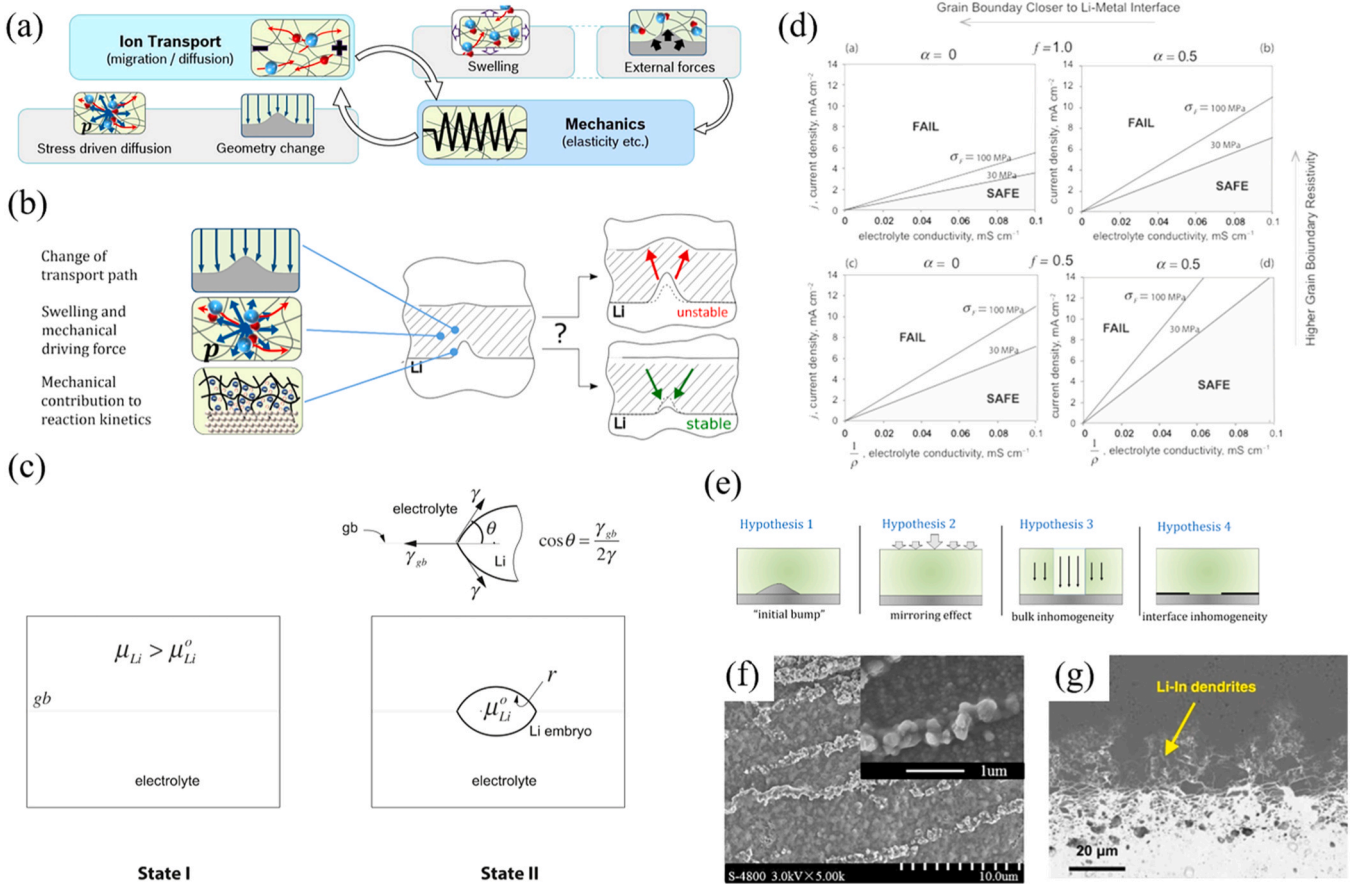


Fig. 7. (a) Relationship between ion transport and mechanics in solid state electrolytes, leading to changes in geometry and stresses [126]. (b) visual summary of electro-mechano-chemical interactions leading to morphological interface stability changes [126]. (c) state I is the supersaturated state, and state II is supersaturation is relieved by the growth of a precipitate. Both states were used to derive the critical current density equation (j^*) [127]. (d) current limit diagram for Solid-State electrolyte - Li interface [127]. (e) four hypotheses for potential trigger mechanisms contributing to inhomogeneous deposition development [126]. (f) SEM imaging of Li dendrite formation with polymer PEO-900000(LiTFSI) electrolyte after 50 cycles [128]. (g) Cross-sectional SEM imaging of Li-In dendrite formation for Li-In|LPSCl|LNO@NCM622 (sulfide electrolyte) cells after 100 cycles [129].

and results in additional protrusion growth that way. Hypothesis 3 suggests that morphological alterations are caused by inhomogeneities in the bulk while hypothesis 4 suggests it is caused by inhomogeneities at the interface between the two materials. Out of these 4 hypotheses, the most significant amount of deposition/ morphological instability was seen to be caused by unwanted alterations to the interface conductivity (hypothesis 4) and resulted in a proportional relationship between the tip velocity and rate of protrusion growth. To further investigate Hypothesis 4, the group of researchers moved forward by altering the Young's modulus of the materials and observing the resulting responses produced by the interface amplitude. Finally, critical current density testing has been performed to analyze the effects of interfacial mechano-chemical problems on the transport properties of solid electrolytes. Depending on the salt-concentration gradient along the separator and presence of external deformations, the current density patterns of the electrode-electrolyte interface will vary [126].

3.4.2. Current status of mechano-chemical characteristics for solid-state electrolytes

As previously indicated, the specific mechanisms and characteristics of mechano-chemical variability in solid state systems can be difficult to define, however, mathematical modeling can provide meaningful approximations to guide battery design and development. In addition to simulation-based modeling, physical observation of dendrite growth and morphological changes in response to mechano-chemical coupling interactions have been observed in polymer electrolyte and sulfide electrolyte systems particularly. While these interactions are relevant in oxide, halide, and hydride systems as well, their specific characteristics have been studied less extensively, likely due to risks associated with the production of hazardous byproducts. Table 6 below summarizes a number of common solid electrolytes alongside their conductivity, stable voltage windows, capacity and number of cycles until loss of performance.

Polymer class solid electrolyte materials, such as polyethylene (PE) and polypropylene (PP), are not capable of inhibiting the growth of lithium dendrites, creating risk for dendrite formation, especially under conditions of high current density [133]. Based on the material composition, degree of crystallinity, and selection of anode and cathode

materials electro-chemo-mechanical stability can vary greatly, however, individual case studies have been performed to provide insight on some of the most common polymer solid electrolytes. An early study by Brissot et al. [134] used microscopy coupled with a CCD camera to observe dendritic lithium deposition over time with respect to selected current density. This baseline study provides a simplistic method for observation of dendrite formation, as well as a preliminary understanding of dendrite growth mechanisms that has been referred to in future studies to further understand mechano-chemical coupling mechanisms in solid electrolyte battery systems.

Sulfide type solid electrolytes introduce risks for the production of decomposition products, such as S, during mechano-chemical degradation, as well as lithium-indium dendrite formation. Sulfide electrode decomposition products can interfere with coulombic efficiency and increase polarization required for initial cycling while dendrite propagation can reduce capacity and lead to eventual cell failure [133]. A 2021 study of the coupling interactions between a $\text{Li}_6\text{PS}_5\text{Cl}$ solid electrolyte and a $\text{Li}(\text{Ni}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2})\text{O}_2$ electrode additionally confirmed the impacts of current and loading on performance of sulfide electrolytes with high current and high loading resulting in thermodynamic and kinetic instability [129]. Overall, the coupling of in situ analyses and mathematical modeling of mechano-chemical interactions can provide significant insight into the performance, stability, and safety of solid-state battery systems to guide further development and improvements.

3.4.3. Potential improvements

Although mechano-chemical interactions provide a significant challenge to solid electrolyte battery development, there are several key methods that have been proposed to minimize the ongoing formation and consequences of dendrites as well as other morphological imperfections. The first possible method is to alter the bulk microscopic properties of the materials using grain size control [135]. Since the grain boundaries of various materials are often prone to dendrite growth, altering the material sizes themselves would be a valid approach to try and combat the issue directly. The second potential method is to minimize interfacial defects as best as possible [135]. As observed in a handful of experiments, both the local and interfacial defects of the

Table 6
Interface level thermal characteristics for solid electrolyte classes.

Electrolyte Category	Interfacial Stability	Morphological Changes	Thermal Runaway	Possible Improvements
Polymer	Acceleration of material decomposition attributed to the open-circuit voltage of the system [110]. Operating voltage dependent stability [121].	Melting shrinkage has been reported as well as deformation risk during combustion.	Not present in all cases (material selection dependent). $\text{Li}_6\text{PS}_5\text{Cl}$ introduces thermal runaway behavior in $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ SPE systems due to reactivity between delithiated $\text{Li}_{1-x}\text{Ni}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ and the solid electrolyte [119].	Addition of vermiculite sheets in the case of $\text{PEO}/\text{LiN}(\text{CF}_3\text{SO}_2)_2$ can address melting shrinkage [122]. Addition of oxides with high thermal stability grade could reduce deformation risks.
Oxide	Largely dependent on the constituent elements and structures of the solid electrolytes being used. Some reported reacting violently with phosphate electrodes under high temperature [112].	Transitions in response to heat observed to cause structural defects (shrinkage, cracking) [110].	Thermal runaway possibly the result of O_2 production causing exothermic reactivity and melting of lithium materials [116]. LAGP-Li, melting of LAGP in argon gas can result in "violent combustion" from O_2 release [123].	Elemental doping treatments (requires further testing). Co-sintering is occasionally used to improve interfacial contact deficits [124].
Sulfide	Likely influenced by cathode state of charge, atmospheric pressure conditions, and charge/discharge cycle number [91].	Shrinkage observed in areas of LPS for $75\text{Li}_2\text{S}-25\text{P}_2\text{S}_5$ glass (LPS)- $\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2$ (NMC) interface during heating [125].	Little work has been done to examine the thermal stability patterns of sulfide solid electrolytes, likely due to the risk of toxic gas production (mainly H_2S). Thermal runaway risk trends require further analysis of this class of materials as the majority of test results are circumstantial. Study of $\text{Li}(\text{CB}_9\text{H}_{10})\text{-Li}(\text{CB}_{11}\text{H}_{12})$ complex hydride solid electrolytes indicated endothermic and exothermic peaks but no largely detrimental runaway patterns [119].	Suppressing exchange of sulfur and oxygen and selecting oxygen stable electrolytes could increase interfacial thermal stability [91]. Development of new methods for stabilization of thermal interfacial degradation byproducts is required [103].
Halide/ Hydride	More complex interactions than other SE types. Current studies are subsequently lacking with regards to testing. Halide interactions without metal ion interactions, such as a $\text{Li}/\text{Li}_2\text{OHCl}/\text{Li}$ cell, have been shown to provide good stability at high working temperatures [118].	$\text{Li}/\text{Li}_2\text{OHCl}/\text{Li}$ thermal processing indicated no pores or cracking while maintaining material thickness. Most require temps between $259\text{--}355^\circ\text{C}$ to cause disorganization of crystallinity [55]. Some exhibit unfavorable structural transformations at low temperatures and failure to recrystallize with sound structure [101].		

reaction/ materials are capable of directly influencing both the rate of ongoing dendrite growth as well as morphological alterations. As a result, attempting to minimize the interfacial defects while maintaining its morphological integrity through methods such as crystal SCE [136] would be extremely beneficial when examining mechano-chemical interactions. The third and final method is to optimize the selections of materials with various processing approaches to try and minimize the amount of stress caused by thermal expansions [137].

4. Fast charging of SSBs and related challenges

Solid-state batteries are becoming increasingly considered for its applications in electric vehicles, pacemakers, and wearable electronics/ devices. However, one of the greatest requirements, yet drawbacks for the current industry is the desire for solid-state batteries to be fast charging and have a high rate of performance [44,46]. This is especially applicable to electric vehicle owners who may find it extremely inconvenient and undesirable to have to wait for an extensive period of charging time [138,139]. Yet, it is difficult to ensure the fast-charging capabilities with using solid-state batteries because fast-charging is commonly known to quickly deteriorate and deplete the battery life [140]. In fact, ongoing capacity degradation of the high-energy-density SSB systems is one of the main factors preventing it from mass production. Some factors that currently contribute to inhibiting fast-charging capabilities of solid-state batteries include low Critical Current Density (CCD), high internal resistance, low/slow ion conduction, ongoing dendrite growth/ high electrical conductivity, poor contact area, increasing thickness of solid electrolytes, low interfacial stability, and low lithium-ion transference number of composite solid electrolytes and a low electrochemical window [41]. While there are ongoing attempts to try and address these factors, some have been more successful than others, there is still a long road ahead to an entirely fast-charging solid-state battery.

4.1. Low critical current density (CCD)

Low charging-discharging current density, specifically low CCD, is one of the main factors preventing solid-state batteries from satisfying commercial needs and applications. More specifically, the power density of solid-state batteries remains at an all time low, when compared to lithium-ion batteries and this means there is not very much capacity available for the entire size of the cell. However, too high of a current density also has its own drawbacks as it has been proven to contribute to decaying a battery's overall capacity. This is directly due to the differences in movement rates between ionic motion and the wire's charge distribution. The wire's charge destructions move at a much faster rate than the ionic motion which takes place within the electrodes/ across the interface and is determined by the diffusion coefficient of the species. Ideally, the diffusion coefficient should be a high number as it directly correlates to how quickly the species can move through a material.

The CCD for lithium plating/stripping in different SSE systems is determined through electrochemical testing, primarily using symmetric Li|SSE|Li cells. The most common method involves cycling the cell at progressively increasing current densities while monitoring the voltage response. The CCD is typically identified as the current density at which a sharp voltage drop occurs, indicating the onset of lithium dendrite formation and cell short-circuiting. However, variations in testing protocols, such as the current step size, duration of each step, total lithium plated/stripped (areal capacity), applied pressure, and temperature, can significantly influence the reported CCD values [141]. Some advanced techniques involve unidirectional plating to avoid complications from stripping-induced void formation, and others focus on minimizing the amount of lithium transferred per cycle to mitigate the impact of void formation on CCD measurements [142]. Standardizing these measurement protocols is crucial for obtaining comparable and reliable CCD

values across different SSE systems, facilitating the development of high-power and long-cycle-life solid-state batteries.

Additionally, attempting to increase the levels of CCD is not a simple task. CCD performance is affected by a number of factors including: mechanical strength, dendrite formation, relative density, poor interface resistance between solid electrolytes and electrodes, ionic and electronic conductivity, as well as electrochemical stability [41]. Since these factors are interconnected and some can be seen to directly affect the fast-charging capabilities of solid-state batteries, it poses as a direct obstacle for the field as changing one thing could negatively affect the other and vice versa. However, this has not stopped researchers and scientists from attempting to move forward and get more answers. For example, Thangadurai and Sarkar [143] suggested a handful of potential improvements that could be carried out to improve the critical current densities of all-solid-state lithium-metal batteries. These strategies [143] include, but are not limited to (1) improving/ modifying the grain boundaries with $\text{Li}_2\text{CO}_3/\text{LiOH}$, (2) creating more uniform interfaces between the Li-metal and SE surfaces to better densify the solid electrolytes and prevent dendrite formation in inorganic solid electrolytes, (3) improving the wettability of solid electrolytes interfaces which can directly reduce interface resistance, (4) lowering the electrical conduction of the bulk electrolyte to ensure the appropriate contact area is maintained, and (5) properly maintaining temperature (which directly affects Lithium wettability on solid electrolytes) and pressure (which can alter the level of resistance at the interface between the Li-metal and solid electrolytes). Overall, Thangadurai and Sarkar [143] suggest that there is no definite solution/ method to improving the CCD of all-solid-state lithium-metal batteries due to its complexity. However, attempting to address the interfacial issues between the cathode and SE is a good starting point as this will ensure better performance of the battery as well as ensure a greater value of CCD for the solid-state battery [143]. Figs. 8 and 9

4.2. Dendrite growth

Ongoing growth of lithium dendrites in solid-state batteries directly correlates to low levels of CCD and therefore further prevents fast-charging capabilities overall. Throughout the years, many factors have been discovered to play a role in contributing to lithium dendrite growth. These factors include low relative density [144], high porosity [144], surface cracks/ bulk defects [145,146], as well as direct formation of the dendrites inside the solid electrolytes [145], rather than simply impaling it from the surface. Additionally, it was confirmed in 2019 by Han et al. that increasing the amount of electronic conductivity in solid-state batteries may be the primary reason for ongoing dendrite growth within solid electrolytes [145].

In their experiment, Han et al. used a time-resolved neutron depth profiling operando technique to look at real-time changes of lithium concentration profiles within three different types of solid electrolytes, lithium phosphorus oxy-nitride (LiPON), $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO), and amorphous Li_3PS_4 (LPS) during the process of lithium plating. Their results insist that lowering the electric conductivity is crucial for the success of solid-state batteries in the long run. However, this poses a direct issue to fast-charging concerns as an increase in electronic conductivity in solid-state batteries is necessary for proper fast-charging capabilities. Although dendrite growth mechanics/ mechano-chemical analysis is an entire topic of its own and has been touched upon in the previous section, its effects are extremely crucial to fast-charging performances of solid-state batteries and is worth mentioning again.

Despite the promise of solid electrolytes in mitigating dendrite formation in SSBs, it remains a predictable challenge due to inherent material properties and electrochemical processes. Quantitative techniques like Electrochemical Impedance Spectroscopy (EIS) track interfacial resistance changes indicative of dendrite growth, while Cyclic Voltammetry (CV) and Chronopotentiometry reveal anomalous voltage/current responses signaling short circuits. In-situ microscopy directly visualizes

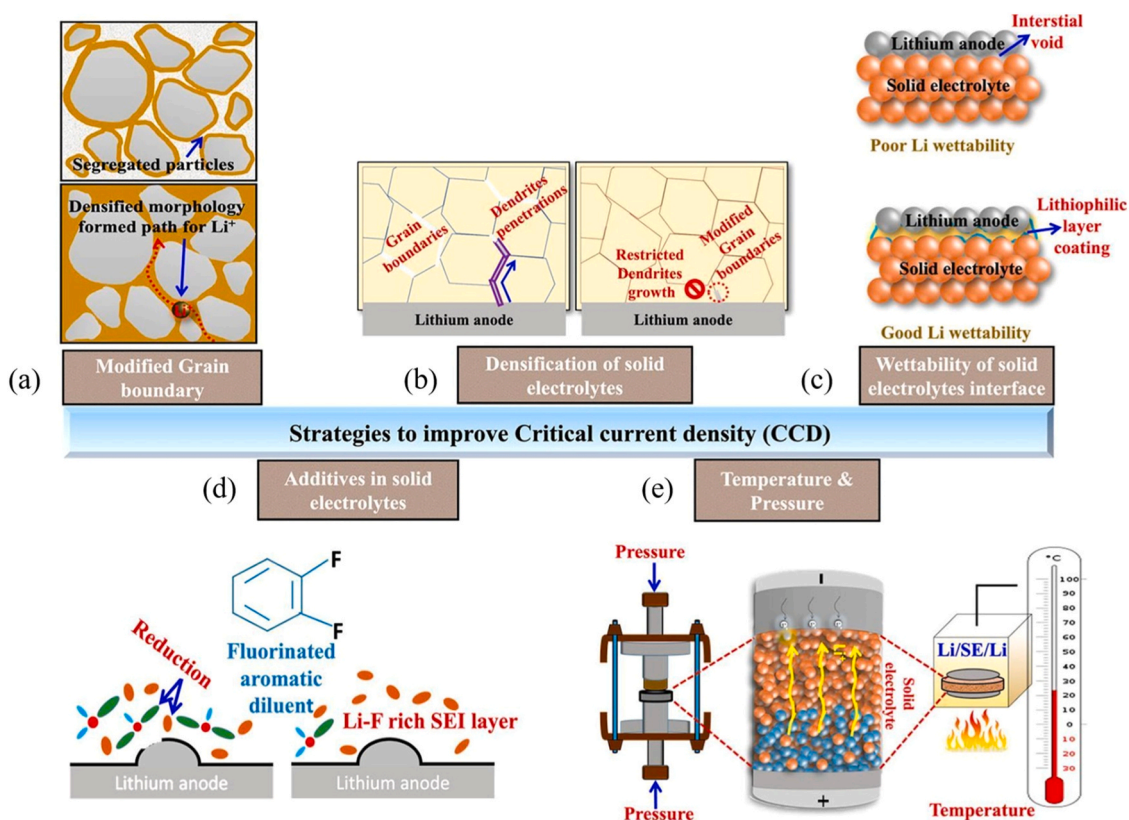


Fig. 8. Methods/ strategies to improving the CCD of all-solid-state lithium-metal batteries [143]. (a) improving/ modifying the grain boundaries with $\text{Li}_2\text{CO}_3/\text{LiOH}$. (b) creating more uniform interfaces between the Li-metal and SE surfaces to better densify the solid electrolytes. (c) Improving the wettability of solid electrolytes interfaces. (d) Lowering the electrical conduction of the bulk electrolyte. (e) Properly maintaining temperature and pressure of the battery.

dendrite nucleation and propagation, allowing for quantitative analysis of their growth kinetics. Nuclear Magnetic Resonance (NMR) and Magnetic Resonance Imaging (MRI) non-destructively quantify dendrite formation. Finally, Finite Element Method (FEM) simulations predict dendrite growth morphology based on electrochemical principles [149–151]. These methods consistently demonstrate predictable patterns of dendrite formation under specific operating conditions.

4.3. High electronic conductivity

Another factor that directly affects the fast-charging capabilities of solid-state batteries and the potential for CCD to go from low to high is the high electrical conductivity rates of the batteries. In fact, high electrical conductivity also leads to the ongoing growth of dendrites and therefore is considered to be a promising issue to address as it can potentially lead to resolving two issues at once. A common technique used to combat the issue of high electrical conductivity is to adjust the thickness of the solid-state electrolytes by reduction. The thickness of the electrolyte is believed to influence both the internal resistance of the battery as well as its concentration polarization [41]. As a result, a smaller thickness is believed to result in both a faster ion conduction and lower internal resistance, especially when comparing the same materials to one another, and is considered ideal for fast-charging applications.

When developing a novel approach to creating thin and easily reproducible electrolyte layers by nanoscopic polymer binding from polysulfide solutions consisting of lithium polysulfide, phosphorus sulfide, and ethylene sulfide, Ping Liu et al. found the room temperature (RT) ionic conductivity to be $2 \times 10^{-5} \text{ S cm}^{-1}$ [147]. Since the ionic conductivity of solid polymer electrolytes is typically $1 \times 10^{-4} \text{ S cm}^{-1}$ [41], this is considered a major drawback to reducing the thickness of the electrolytes because it seems to also reduce RT ionic conductivity in an attempt to reduce electrical conductivity and dendrite formation. In

fact, reducing the value of RT ionic conductivity results in a decrease in stability and therefore makes the solid-state battery undesirable for applications. In another experiment, researchers attempted to adjust the solid electrolyte thickness by creating Ta-LLZO powders through solid-state reaction. In general, Ta-LLZO is more often favored over other powders such as Al-LLZO and Nb-LLZO because it has better stabilization potentials and can increase ionic conductivity levels with Li metals better than others [152]. More specifically, once the Ta-LLZO powders in the experiment were developed, it was then followed by a high-temperature sintering process and were finally pressed into green pellets under gradual and static pressure. This resulted in the formation of thinner LLZO solid-state electrolytes ranging from 800 to 1000 μm and 10–20 mm in diameter [41]. This is a significant amount of reduction considering the typical thickness for an inorganic SSE is around several hundreds of micrometers [41]. However, this reduction in thickness also led to a decrease in the amount of production by manufacturing overall. This was due to the ceramic material being inherently more brittle than others [153] and as a result, became a major drawback for the technique. Furthermore, it puts scientists and researchers in a position where they are required to either prioritize large scale manufacturing or fast-charging capabilities of solid-state batteries, but not necessarily both simultaneously.

4.4. Low interfacial stability, high internal resistance, and low ionic conductivity

Low interfacial stability is another factor that contributes to low CCD levels, thus further preventing fast-charging capabilities of solid-state batteries in the long run. Interface-level thermal stability refers to the contact/ interaction between the solid-state electrolytes and electrodes (cathode and anode). This means that the interface relationship is capable of directly interfering with the ionic conductivity of the system

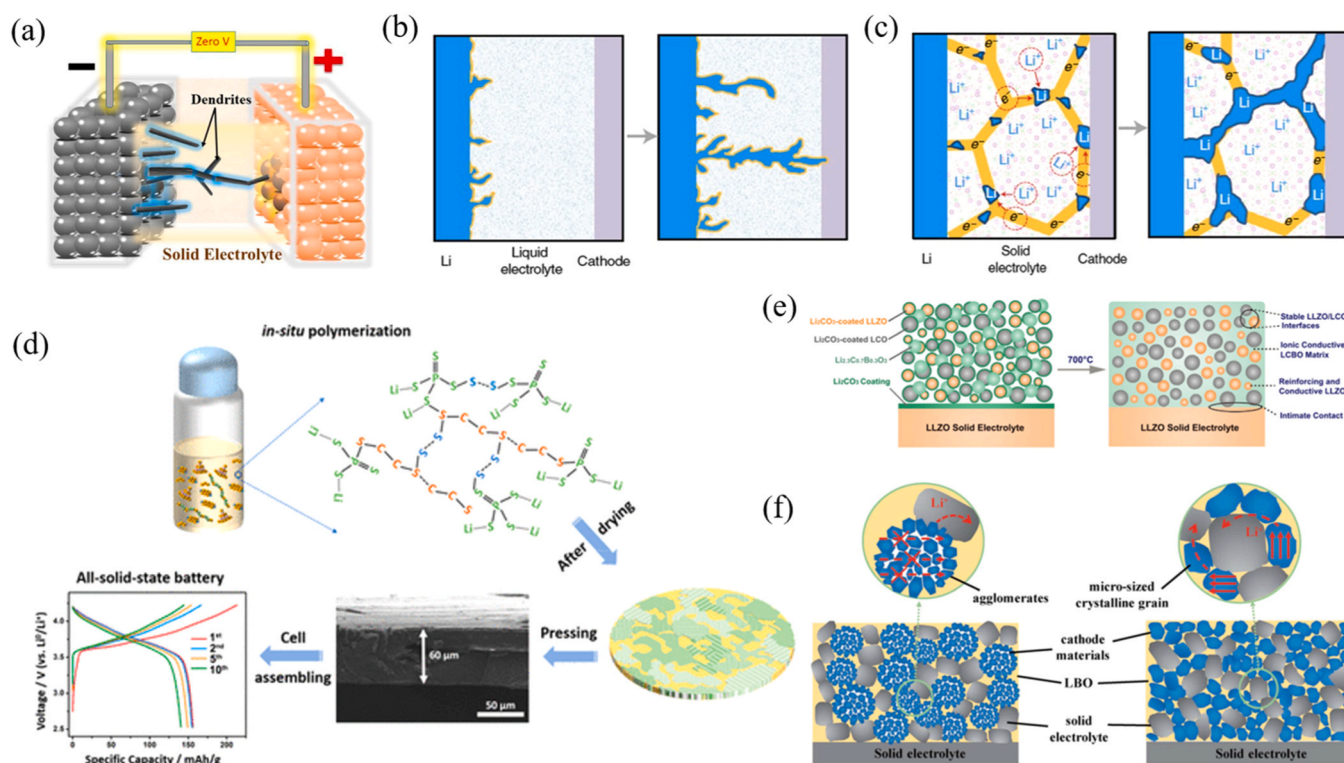


Fig. 9. (a) Ongoing lithium dendrite formation within solid-state batteries is one of the main issues preventing SSBs from commercialization [143]. (b) the ongoing formation growth of lithium dendrite in traditional liquid LIBs [41]. (c) lithium metal nucleation on grain boundaries, followed by lithium filament penetration inside solid electrolytes [41]. (d) novel synthesis approach for nanocomposite solid electrolyte-in-polymer matrices (SEPM) based on *in situ* polymerization reaction [147]. (e) a novel experiment to avoid and remove impurity phase production by high temperature co-sintering [148]. (f) the schematics of all-ceramic interfaces between the cathode and electrolyte [148].

as well. As a matter of fact, low, poor, and unstable interfacial contact between the solid electrolytes and electrodes means that there is an undesirably large interfacial resistance, thus resulting in low CCD and directly hindering fast-charging capabilities of solid-state batteries. Since the ionic conductivity of solid-state electrolytes (ion transport in electrolytes) and the interface resistance (ion transport between electrodes and electrolytes) are two chemical processes that play a significant role in affecting a battery's charging behavior, it would be ideal to consider increasing the ionic conductivity while simultaneously lowering the levels of internal resistance to achieve fast-charging capabilities, since these two factors can lead to morphological changes and decomposition of a battery. One technique often used to improve interfacial deficits is known as high temperature co-sintering. However, a common drawback to this technique is that it triggers unwanted material decomposition, increasing the production of high-impedance impurities, and overall weakening of the electrochemical capabilities. Thermal gradients play a crucial role in triggering localized decomposition by creating non-uniform temperature distributions within a material or system. Since the rate of thermal decomposition reactions is highly temperature-dependent (often following the Arrhenius equation), regions experiencing higher temperatures will undergo decomposition at a significantly faster rate than cooler areas [154]. This disparity in reaction rates leads to a localized breakdown of the material, concentrating the decomposition process in the hotter zones while leaving the colder regions relatively intact. This phenomenon is particularly important in systems where heat transfer is inefficient or where localized heat sources exist, leading to the development of steep temperature gradients that can exacerbate material degradation in specific areas [155]. This technique also results in high interfacial resistance due to the chemical instability of cathode materials and solid electrolytes, thus directly resulting in low CCD and therefore takes us back to the original issue [148]. Finally, in an attempt to address this

issue, Han et al. [145] carried out a reaction that formed an intermediate phase to remove the potential impurities produced by high temperature sintering of LCO and LLZO. More specifically, the experiment relied on a welding agent ($\text{Li}_{2.3}\text{Co}_{0.7}\text{B}_{0.3}\text{O}_3$) that welded the LCO and LLZO together and took place at 700 °C, which is way lower than conventional sintering taking place at 1100 °C [148].

Doping strategies using elements like aluminum and gallium significantly influence the Li-ion conductivity and phase stability of garnet electrolytes (e.g., $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$) under fast charging conditions. Doping with Al^{3+} or Ga^{3+} on the Li^+ site generally stabilizes the high Li-ion conductive cubic phase of the garnet structure at room temperature, preventing the formation of the low conductive tetragonal phase [156]. This stabilization is achieved by creating Li vacancies to maintain charge neutrality, which in turn enhances Li-ion mobility and thus, the overall ionic conductivity crucial for fast charging. Furthermore, doping can improve the grain boundary conductivity by reducing impedance and enhancing the density of the electrolyte, facilitating faster Li-ion transport. The specific concentration of the dopant is critical, as excessive doping can lead to the formation of secondary phases that impede conductivity and potentially affect the long-term phase stability, especially under the stress induced by rapid Li intercalation and deintercalation during fast charging [157].

4.5. Low electrochemical window

The electrochemical stability window of solid electrolytes is typically low and is not ideal for fast-charging capabilities. More specifically, a typical voltage window often compensates for the fast-charging efforts by exerting overpotential on the entire solid-state battery. This then leads to an increase in voltage, thus going above the stable voltage window and is considered to be a major drawback of the fast-charging feature. In fact, when the electrochemical window is incapable of

reaching the working voltage range of the electrodes in the system, this results in unwanted electrochemical reactions at the interface and leads to the formation of undesirable by-products caused by unexpected oxidation and reduction reactions to the electrolytes themselves [41]. For example, the electrochemical stability window of the solid electrolyte $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS) has been a subject of investigation, with early reports suggesting a wide window of 0–5 V based on conventional measurements using Li/LGPS/inert metal cells [158]. However, these methods often overestimate the true electrochemical stability window due to limited contact area. First-principles calculations and experimental methods employing Li/LGPS/electrolyte-carbon cells have revealed a significantly narrower intrinsic electrochemical stability window for LGPS [158]. These studies indicate that LGPS is thermodynamically unstable against lithium metal, leading to decomposition at low potentials, and also undergoes oxidation at higher potentials, making the practical electrochemical stability window considerably smaller than initially reported. Overall, this issue is considered to be a major cause of both large interfacial resistance and rapid capacity decay [41]. In an attempt to combat this issue, Pan et al. [159] decided to increase the electrochemical stability window of solid electrolytes by making this value greater/ wider while also simultaneously inhibiting the ongoing formation of lithium dendrites [41]. More specifically, they created a new quasi-double-layer electrolyte that is capable of exhibiting various surface characteristics when coming into contact with the electrodes (either cathode or anode) [159]. This technique therefore eliminates all additional and unwanted electrolyte interfacial resistance.

4.6. Future directions for achieving fast-charging capabilities

Overall, achieving fast-charging capability of solid-state batteries is not an easy task and many factors must be considered as one issue may be interrelated to the other and changing one thing could lead to a cascade of unwanted outcomes. As a result, it is detrimental to determine and prioritize some factors over others prior to developing potential solutions to the current fast-charging issues for solid-state batteries as it is extremely difficult to meet the needs of all ongoing issues/ demands.

Thangadurai and Sarkar [143] suggested five potential improvements that could be carried out to improve the critical current densities of all-solid-state lithium-metal batteries. These strategies include (1) improving/ modifying the grain boundaries with $\text{Li}_2\text{CO}_3/\text{LiOH}$; (2) creating more uniform interfaces between the Li-metal and SE surfaces to better densify the solid electrolytes; (3) improving the wettability of solid electrolytes interfaces; (4) lowering the electrical conduction of the bulk electrolyte; and (5) properly maintaining temperature and pressure of the battery. However, Thangadurai and Sarkar [143] ultimately concluded that there is no definite method to improving the CCD of all-solid-state lithium-metal batteries. However, attempting to address the interfacial issues between the cathode and SE ensures better performance of the battery as well as ensure a greater value of CCD for the solid-state battery in the long run [143]. Similarly, the ongoing growth of lithium dendrites in solid-state batteries directly correlates to low levels of CCD and there is a common factor researchers look at when attempting to achieve fast-charging capabilities of solid-state batteries. When Han et al. used a time-resolved neutron depth profiling operando technique to look at real-time changes of lithium concentration profiles within three different types of solid electrolytes, their results showed that lowering the electric conductivity is crucial for the success of solid-state batteries in the long run [145]. However, this poses a direct issue to fast-charging concerns as an increase in electronic conductivity in solid-state batteries is necessary for proper fast-charging capabilities. Similarly, the high electrical conductivity rates of the batteries is another factor that directly affects the fast-charging capabilities of solid-state batteries and the potential for CCD to go from low to high. When attempting to address the issue of high electrical conductivity, adjusting the thickness of the solid-state electrolytes by reduction. Yet, a

major drawback to this technique is that it results in the reduction of RT ionic conductivity as well. Additionally, the reduction in thickness results in the ceramic material being inherently more brittle than others [153] and therefore seems to lead to a decrease in the amount of production by manufacturing overall, thus becoming a major drawback for the technique.

Low, poor, and unstable interfacial contact between the solid electrolytes and electrodes means that there is an undesirably large interfacial resistance, thus resulting in low CCD and directly hindering fast-charging capabilities of solid-state batteries. A common technique often used to improve interfacial deficits is known as high temperature co-sintering. However, a common drawback is that it triggers unwanted material decomposition, increasing the production of high-impedance impurities, and overall weakening of the electrochemical capabilities. This technique also results in high interfacial resistance due to the chemical instability of cathode materials and solid electrolytes, thus directly resulting in low CCD and therefore takes us back to the original issue [148]. Finally, the electrochemical stability window of solid electrolytes is typically low and is not considered ideal for fast-charging capabilities. Thus, Pan et al. [159] decided to increase the electrochemical stability window of solid electrolytes by making this value greater/ wider while also simultaneously inhibiting the ongoing formation of lithium dendrites [41]. Their technique resulted in the elimination of all additional and unwanted electrolyte interfacial resistance and was considered to be a major success. Although the common issues that affect fast-charging capabilities have just been laid out again, these barely scratch the surface and the complexity of these issues prove that achieving fast-charging capability of solid-state batteries is not an easy task. However, resolving all these ongoing issues should still be considered the end goal and would guarantee 100 % fast-charging capabilities for all existing solid-state batteries, if properly achieved.

5. Operando analysis for in-situ characterization

5.1. Approaches and methodology

There are two technique approaches to obtaining essential information on interphases in SSBs. One of the approaches is a combination of a suitable sample preparation procedure or the use of modified conventional batteries with the subsequent application of an analytical technique, usually a surface analytical technique, to gather structural and/or chemical data. This technique is known as the uncovering approach technique [160]. The other approach is the bulk technique analysis. Bulk technique analysis is defined as the direct use of analytical techniques featuring a large sampling depth (SD) on SSBs [160]. These approaches have three distinct analysis techniques to unveil the chemo-electromechanical phenomena at interfaces created by batter charge and discharge [160]. These three different analytical techniques are ex-situ, in situ, and operando analysis. All three are defined below. Ex-situ analysis is defined as the analysis carried out on a sample prepared from a battery that has been subjected to an electrochemical input and subsequently disassembled [160]. In situ analysis is defined as the battery, which has been previously modified or treated with one of the aforementioned sample preparation procedures, being introduced inside the analysis chamber or special cell of the instrument. The electrochemical input can then be applied to the battery in this place, but the analysis of the interphase is only carried out after the interruption of the electrochemical input. Switching between electrochemical input and analytical measurement allows for monitoring the evolution of the interphase [160]. Operando analysis is similar to in situ analysis, but the probing of the interphase is performed simultaneously with the application of the electrochemical input. This operation mode enables the evolution of the interphase to be followed in real-time. Given the two approaches offer various techniques associated with different significance, the technique's applications and significance will only focus on the most important experiments for SSBs [160].

5.2. The uncovering technique approach

The uncovering technique approach is defined as a combination of a suitable sample preparation procedure or the use of modified conventional batteries with the subsequent application of an analytical technique, usually a surface analytical technique, to gather structural and/or chemical data [160]. In the uncovering approach, the strategy is to gain access and probe the buried interfaces in solid-state batteries. This strategy requires two steps: (1) expose the surface of diverse materials in the contained electrochemical cell and (2) analyze the treated samples with a surface technique [160]. The techniques that use the uncovering approach include Optical microscopy (OM), Scanning Electron Microscopy (SEM), Energy-Dispersive X-Ray Spectroscopy (EDS), Transmission Electron Microscopy (TEM), Scanning Transmission Electron Microscopy (STEM), Auger Electron Spectroscopy (AES), Scanning Auger Microscopy (SAM), Raman Spectroscopy and Microscopy, X-Ray Photoelectron spectroscopy (XPS), Scanning Probe Microscopy (SPM), X-Ray Absorption (sXAS), Time-Of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS) [1]. These techniques perform all three analytical techniques: ex-situ, in-situ, and operando analysis. For the sake of simplicity, only optical microscopy (OM) and scanning probe microscopy (SPM) due to their significance in solid-state batteries.

5.3. Optical microscopy (OM)

Optical microscopy mainly performs operando analysis in solid-state batteries. Akitoshi Hayashi, a chemistry professor from Osaka Prefecture University, provides an example of optical microscopy performance. Hayashi used optical microscopy (OM) to study the limitation of graphite electrodes in graphite/LPS/InLi batteries [161]. Optical mapping of a cross-section of the cell showed a gradual decrease in the degree of lithiation in the whole electrode after 4 cycles; however, the loss of capacity in Li insertion was more severe at the current collector/solid electrolyte interface. This is a clear usage of operando measurements on SSB.

5.4. Scanning probe microscopy (SPM)

Scanning Probe Microscopy (SPM) contributes to both ex-situ and in-situ/operando studies. The ex-situ analysis of cross-sections in solid-state batteries is predominantly performed using scanning electron microscopy and is widely recognized as the most widely employed technique for this purpose [160]. Gerbrand Ceder, a Distinguished Professor of Materials Science and Engineering at the University of California, Berkeley, conducted an ex-situ analysis to examine the microstructure evolution of a composite cathode in a $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$ (NMC 532)/75 $\text{Li}_2\text{S}_2\text{P}_2\text{S}_5$ (LPS)/InLi battery [162]. Ceder and his research team studied the battery's internal contents, including solid electrolytes, conductive additives, and voids. Using back-scattered electrons (BSEs) calculation, Ceder discovered an increase in void volume after 50 cycles, leading to reduced surface contact between the active material and other solid-state components of the cathode. This decrease in contact was found to be associated with capacity fading. Consequently, Ceder and his team concluded that the mechanical degradation of electrodes is primarily caused by the loss of solid-solid interfaces, posing a significant obstacle in the development of efficient solid-state batteries. This was a good usage of ex-situ analysis with scanning probe microscopy.

Similar to ex-situ studies, in-situ studies also provide extensive analysis of cross-sections in solid-state batteries. Shirin Kaboli, a researcher from Hydro-Quebec, a power utility company responsible for electric distribution in Quebec, conducted an in-situ study to investigate the cross-sections of solid-state batteries (specifically containing PEO: LiTFSI solid electrolyte) during cycling [163]. This comprehensive analysis involved examining the growth of dendrites at the Li/solid electrolyte interface. The study revealed the presence of carbon, oxygen, and lithium within the dendritic structures, suggesting that they were

composed of lithium carbide, which likely originated from polymer reduction. This was a good example of in-situ studies with scanning probe microscopy.

5.5. Bulk approaches in characterization

Other potential techniques of characterization would involve bulk type approaches. Similar to its counterpart, the surface analysis technique, the bulk approach also provides some analytical techniques for solid-state batteries. However, the role of the bulk approach for SSBs is to provide a study of buried interfaces in SSBs providing a larger penetration depth. This allows the investigation of some materials in the SSB cells without the need for disassembling, cutting, or ion milling procedures. The common bulk approach methods are listed in no exact order: neutron depth profiling (NDP), micro- and nano-X-ray computed tomography (X-CT), and nuclear magnetic resonance (NMR). This approach mainly uses ex-situ; however, the uses of in-situ and operando approach isn't uncommon. The section will now digest each technique and its performance approach beginning with Neutron Depth Profiling.

5.6. Neutron depth profiling (NDP)

Neutron Depth Profiling, or NDP, is a commonly used method in the bulk approach. Refer to the following equation below:



In Eq. 11, a low energy neutron (n), interacts with a ${}^6\text{Li}$ atom which produces an α (${}^4\text{He}$) and a triton (${}^3\text{H}$) particle [160]. ${}^4\text{He}$ and ${}^3\text{H}$ particles are released with energies of 2055 and 2727 keV as a result of the first collision. In addition, the two ejected particles suffer from collisions thus decreasing their original energies at the moment of their formation. The energy loss explains the initial location of the ${}^6\text{Li}$ atom inside the SSBs, therefore, the high energy decrease involves a deeper position of the buried lithium atom inside the staking material. This is what happens in Neutron Depth Profiling. This technique provides an analysis of SSBs which must contain uniform materials and flat surfaces found in micro batteries. The sampling depth is about $< 50 \mu\text{m}$. NDP measurements on solid-state batteries are performed within the in situ or operando approach. The NDP method has evidently shown to be important in the evaluation of Li transport dynamics, particularly at interfaces. This is seen during consecutive plating and stripping cycles in Li/solid electrolyte/Li symmetrical cells. The formation and growth of Li dendrites at interfaces are particularly looked at.

Wang et al. found in in situ experiments that detrimental Li plating can occur inside the bulk solid electrolyte instead of the Li electrode/solid electrolyte interface in $\text{Li/Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}/\text{Cu}$ and $\text{Li/Li}_3\text{PS}_4/\text{Pt}$ solid-state cells [145]. Wang's group found that the growth of Li deposits results in the failure of the battery because of the short circuit. The group further determined that the Li plating inside the electrolyte is attributed to the residual but no negligible electronic conduction of $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ and Li_3PS_4 . However, LCO/LiPON/Cu micro batteries had no evidence of Li deposition in the solid electrolyte given LiPON has very low ionic conductivity. They suggested that the electronic conductivity of solid electrolytes is an essential parameter for the life cycle of the battery. The operando approach has been used in an optical microscopy experiment on an NMC622/LPS/Li battery: the study was conducted by Minghao Sun et. al, who found that Li plating inside the solid-state electrolyte occurs before the growth of dendrites at the Li/LPS interface [164].

5.7. Micro- and nano- X-ray computed tomography (X-CT)

Micro- and Nano-X-Ray Computed Tomography, also known as X-CT, is an alternative method used to visualize and evaluate the microstructure of materials in SSBs [160]. This technique is considered a pretty useful technique in order to digest the electromechanical dynamics of

materials and interphases in SSBs. The advantage of this technique is that batteries comprising composite electrodes and solid electrolytes can be analyzed given it has been less looked at than other SSBs phenomena. The method can be both approached with ex-situ and in situ/operando.

The purpose of ex-situ experiments has ranged from analyzing the electrochemical performance and stability of devices to dedication to research on symmetric Li/solid electrolyte/Li cells. For instance, research Nitash Balsara formed a synchrotron-based X-CT experiment aimed at elucidating and mastering the growth of Li dendrites and protrusions in symmetric cells containing a rigid polystyrene-block-poly (ethylene oxide) (PS-b-PEO) block copolymer blended with LiTFSI salt [160,165–169]. Balsara observed that Li protrusions formed from crystalline impurities inside the Li metal electrode. The formation of dendrites indicates uneven Li deposition at Li electrode/solid electrolyte interfaces. The results suggest Li protrusions and dendrites start in crystalline impurities such as electrical insulators, initially buried in the bulk of Li electrode, beneath the electrode/solid electrolyte surface. This finding is essential as efficient suppression of dendritic structure is only possible with Li electrodes with high purity.

In situ and operando experiments were also reported for Micro- and Nano-X-Ray Computed Tomography. For instance, in-situ experiments, performed by M.T. McDowell et. al., were used to evaluate the evolution microstructure of the solid electrolyte in a $\text{Li}/\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3/\text{Li}$ symmetric cell [170]. McDowell's group observed a formation and growth of an interphase layer between the electrodes and solid electrolyte and found that the failure of the cell was due to the following: formation of microcracks, propagation of microcracks, and widening of microcracks. Operando experiments' contribution to X-CT research involved Yuta Kimura et. al.'s group. Kimura et. al.'s group published their report on an operando experiment where the group combined synchrotron X-CT and X-Ray absorption near edge structure (XANES) spectroscopy to evaluate operando the 3D spatial distribution of the state of (dis)charge (SOC) of LCO particles in a composite electrode containing $\text{Li}_{2.2}\text{Co}_{0.8}\text{B}_{0.2}\text{O}_3$ (LCBO) as a solid electrolyte without the presence of conductive additives [160,171,172]. The authors found that a heterogeneous distribution of SOC was formed and the SOC maps after charge and discharge pointed to a less efficient de-lithiation or lithiation. So, the LCO particles were embedded in the center of agglomerates of active material particles. The group concluded that poor ionic conductivity at LCO/LCO particle interfaces may be the reason for the behavior [161].

5.8. Nuclear magnetic resonance (NMR)

Nuclear Magnetic Resonance, or NMR, is used to assess the diffusion and distribution of Li^+ and Li metal at interfaces. For SSBs, NMR is usually carried out in the solid state, thus making this method an essential tool [160]. It is worth mentioning that NMR measurements have only been performed ex situ experiments. An important parameter for the performance of the battery relies on the rate of spontaneous exchange of Li^+ at solid/solid interfaces [160].

Ex Situ Experimentation has several uses in solid-state NMR applications to study the interfaces in SSBs. One of which is the tracking of the formation and growth of Li dendrites by ^7Li magnetic resonance imaging (MRI). Clare Grey et al. study this technique in order to visualize and detect the onset and evolution of Li microstructures in a $\text{Li}/\text{Li}_{6.5}\text{La}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$ (LLZTO)/Li symmetric cell [173]. The stark difference in chemical profiles between ^7Li environments in the bulk metal and LLZTO allowed the unambiguous localization of each phase, including dendritic Li, in the device. Importantly, the experiments revealed the growth of Li microstructures on the surface of the Li metal electrodes, probably corresponding to dendrites, before any sign of degradation in the voltage–time curves during the galvanostatic cycling of the system. Furthermore, it was shown that the growth of dendrites occurred simultaneously on the plated and stripped lithium electrodes. The unexpected formation of Li microstructures on the stripped

electrode was ascribed to roughing of the surface due to uneven Li release and subsequent formation of hot spots exhibiting enhanced current densities. Below are tables providing a brief recap summary of both the uncovering and bulk approach techniques. The tables also provide what studies each technique used and the technique's significance (Tables 7 and 8). The studies have been color-coded for visual simplicity. A more detailed summary of the discussed techniques and studies can be read in Lopez's article [160]. Table 9

5.9. Operando X-ray diffraction (XRD)

Operando XRD is a powerful analytical technique used to study the structural changes in battery electrodes during cycling. This method provides real-time insights into the phase transitions and crystal structure modifications that occur in electrode materials, which are crucial for understanding and improving the performance of solid-state lithium batteries.

Operando XRD allows for the observation of phase transitions and structural changes in electrode materials during battery operation. This technique can capture the dynamics of lithium diffusion and the associated structural transformations in real-time, providing valuable insights into the electrochemical mechanisms at play. [219–221]

Studies using operando XRD have shown that NMC cathodes undergo significant structural changes during fast discharging, with reaction inhomogeneity observed across the electrode. The technique has been used to correlate these changes with the state of charge and discharge rates, providing insights into the design of electrodes for improved performance [220,222].

In blended positive electrodes, operando XRD has been used to study the charge transfer dynamics between different active materials. The findings highlight the impact of voltage profiles and state of charge on the distribution of current and charge transfer between components, which is crucial for optimizing electrode design [223].

Operando XRD has revealed significant inhomogeneities in lithium distribution across electrodes, which can affect battery performance and longevity. Addressing these inhomogeneities through improved electrode design and material selection is a key area for future research [224,225].

High-voltage cycling of NMC cathodes leads to increased structural distortion and transition metal dissolution, contributing to capacity fade. Understanding these effects through operando XRD can inform strategies to mitigate degradation and enhance battery life [222].

Operando XRD is an essential tool for advancing the understanding of structural dynamics in solid-state lithium batteries. By providing real-time insights into the phase transitions and structural changes in electrode materials, this technique supports the development of more efficient and durable battery systems. Future research should focus on addressing the challenges of inhomogeneity and degradation to further improve battery performance.

6. Summary and future directions

The advancements delineated within this review of SSB technology herald a paradigm shift in energy storage solutions. To fully capitalize on the potential of SSBs, concerted efforts in several strategic research areas and technological innovations are imperative:

6.1. New generation materials and novel properties

Ongoing progress in developing new classes of materials plays a quintessential role in the evolution of SSB technologies. Sustained and rigorous research initiatives are imperative for the synthesis and characterization of innovative electrode and solid electrolyte materials, which are the cornerstone of enhancing the overall performance and reliability of SSBs. One of the primary objectives in SSB material development is to augment ionic conductivities within solid electrolytes.

Table 7

Ionic conductivities and electrochemical stability Windows of modified solid electrolytes and battery performance of high-voltage all solid-state lithium-ion batteries [133].

Solid electrolyte	Conductivity (S cm^{-1})	Stable window (V)	Cell constitution	Capacity (mA h g^{-1})	Cycle performance (%)
PEO + LiTFSI + wheat flour electrolyte	2.62×10^{-5}	—	NCM811/Li	133 (0.1 C)	47.30
PMA + PEO	2.05×10^{-4}	—	LiCO_2/Li	119 (0.1 C)	91.20
PAN + PAN@LAGP + PEGDA	3.7×10^{-4}	0–5 V	NCM622/NCM811/Li	185 (0.5 C)	81.50
PEO + SN + LiTFPB + LiTFSI	0.5×10^{-3}	0–5 V	LiCO_2/Li	152 (0.1 C)	83.50
P(PO/EM) + LiTFPB	1.55×10^{-4}	—	$\text{LiFe}_{0.2}\text{Mn}_{0.8}\text{PO}_4/\text{Li}$	141.1 (0.1 C)	92.10
PS/PEG/PS	1.1×10^{-3}	0–4.5 V	NCM523/Li	130 (0.1 C)	96.10
Ca–CeO ₂ /PEO	1.3×10^{-4}	0–4.5 V	LiCO_2/Li	113 (0.1 C)	70.70
LLZTO/PVDF	2.73×10^{-4}	0–4.77 V	LiCO_2/Li	120 (0.1 C)	94.10
PAN- <i>in situ</i>	3.5×10^{-4}	—	NCM622/Li	177.3 (0.1 C)	93.70
PVEC–LiTFSI–SiO ₂	1.35×10^{-3}	3.0–4.6 V	LiCO_2/Li	153 (0.1 C)	94
PVA-g-PCA-IL-HT	4.78×10^{-3}	0–5.17 V	LiCO_2/Li	188.7 (0.2 C)	97
PME–LiPVFM–LiTFSI	3.57×10^{-4}	0–5 V	LiCO_2/Li	196.3 (0.1 C)	91.90
PPC–PCDF–LLZTO–LiTFSI	1.3×10^{-4}	0–5.1 V	NCM811/Li	200 (1 C)	82.80
PTFE + Li ₆ PS ₅ Cl + Li ₃ InCl ₆ + LLZTO	5.2×10^{-4}	—	NCM811/Li	173.6 (1/3 C)	94.40
Li _{3-x} Yb _{1-x} Zr _x Cl ₆	1.1×10^{-4}	—	NCM622/Li	170.6 (0.2 C)	80
PDOL + LiDFOB	2.46×10^{-4}	0–4.9 V	NCM811/Li	175 (0.2 C)	—
PVAC + TAGDA + AIBN	1.02×10^{-4}	0–5.2 V	NCM811/Li	200 (0.1 C)	90
PSiOM	0.41×10^{-4}	0–4.8 V	NCM622/Li	138 (0.1 C)	90
PEO + LiTFSI + LiI	2.1×10^{-4}	—	NCM811/Li	125 (0.5 C)	80
PVEC	2.08×10^{-3}	0–4.8 V	LiCO_2/Li	140 (1 C)	89

Research should focus on the discovery and optimization of materials that facilitate faster ion transport while maintaining electrochemical stability across the battery's operational voltage window. This includes exploring lattice structures that allow easier movement of ions, potentially involving the doping of materials with atoms that can create more mobile ion pathways.

Concurrently, it is critical to enhance the mechanical stability of these materials to prevent structural degradations such as cracking or delamination under mechanical stress or thermal fluctuations. This involves engineering materials with robust mechanical properties that can withstand the stresses induced by ionic movement and volume changes during charge-discharge cycles. Research in composite materials or hybrid structures combining polymers and ceramics could provide pathways to achieve these characteristics. Another pivotal aspect of material science research is the operational integrity of SSBs under a diverse array of environmental conditions. Materials must be developed that not only perform optimally in standard conditions but also retain their functional characteristics under extreme temperatures, pressures, or humidity levels. This is particularly relevant for applications ranging from electric vehicles, which require operation in varying climates, to portable electronics, which may encounter various environmental stressors. Additionally, enhancing the interface stability between electrodes and solid electrolytes is crucial for maintaining long-term battery performance and safety. The interfacial region often presents a significant challenge due to the potential for chemical reactivity and physical mismatches in properties such as thermal expansion coefficients. Advanced surface engineering techniques, including atomic layer deposition (ALD) or surface coating strategies, could be explored to create stable, compatible interfaces that mitigate degradation mechanisms.

By addressing these key areas, the scientific community can substantially advance the material science surrounding solid-state batteries, paving the way for more efficient, durable, and versatile energy storage solutions. These advancements will not only improve the technical specifications of SSBs but also enhance their commercial viability, driving forward their adoption across a broad spectrum of applications.

6.2. Scalability of manufacturing processes

The advancement from laboratory prototypes to industrial-scale production of SSBs encapsulates a critical phase fraught with technological and engineering challenges. These challenges underscore the need for robust methodologies to uphold the integrity and performance

of SSBs during large-scale manufacturing processes. Systematic investigations and innovations are required to establish scalable manufacturing protocols that deliver consistent quality and economically viable production rates. A Key to the transition of SSBs to mass production is the development of synthesis methods that can be effectively scaled without compromising the material properties critical to battery performance. This includes adopting chemical and physical vapor deposition techniques, sol-gel processes, and advanced ceramic processing technologies adaptable to large volumes and high throughput.

The intricate architecture of SSBs requires precise assembly techniques to maintain the alignment and uniformity of electrode and electrolyte layers. Advanced robotic and automated systems could enhance the precision of stacking processes, thereby reducing variability and improving the reproducibility of battery cells. Implementing rigorous inline quality control systems is vital to achieve reproducibility and sustainability. Inline systems would utilize advanced sensing and imaging technologies to monitor the integrity and consistency of materials and components throughout the manufacturing process. Real-time diagnostics coupled with automated feedback mechanisms would ensure the detection and correction of production anomalies, thereby enhancing the yield and reliability of the final products.

To achieve a competitive market positioning, developing production strategies that leverage economies of scale is essential. This involves optimizing the material supply chain, reducing energy consumption during manufacturing, and minimizing waste through closed-loop recycling processes. Additionally, partnerships with raw material suppliers and technology developers can help reduce costs and secure supply chains. Manufacturing processes must also adhere to stringent regulatory standards and environmental guidelines to promote sustainability and reduce ecological impacts. This includes the implementation of cleaner production techniques, the use of non-toxic and recyclable materials, and the adherence to international standards on emissions and waste management.

Focusing on these strategic areas can significantly enhance SSBs' scalability and manufacturability, paving the way for their widespread adoption and integration into global markets. The continuous evolution of manufacturing technologies and processes will be instrumental in overcoming the existing barriers to large-scale production, catalyzing SSB technology's transition from niche applications to mainstream energy solutions.

Table 8
Common uncovering approach techniques.

Techniques	Studies	Significance
Optical Microscopy (OM)	Operando	Operando: Studied of lithiation of graphite electrode in a graphite/LPS/InLiX battery [161]. Analyzed cross-section of S ₈ /PEO:LiTFSI (20:1)-LLZTO/Li SSB [3].
Scanning Electron Microscopy (SEM) and Energy-Dispersive X-ray Spectroscopy (EDS)	Ex-situ/In-situ/Operando	Ex-situ: Visualized cross sections prepared by FIB [162,174–176], BIB polishing [177, 178], mechanical cutting [179]. In-situ/Operando: SEM analysis of diverse cycling SSBs containing PEO:LiTFSI solid electrolyte [163,180–183]. Possibility of visualizing real-time variation in thicknesses of 3 stack materials (composite cathode, solid electrolyte and Li anode) [163].
Transmission Electron Microscopy (TEM)/Scanning Transmission Electron Microscopy (STEM) and Associated Techniques	Ex-situ/In-situ/Operando For In-situ and Operando: Based on TEM/STEM	Ex-situ: Prepared thin cross sections for robust/reproducible TEM/STEM studies [184–186]. Created LiCoO ₂ /LIPON/Si nanobattery by operation of FIB on the corresponding micro battery [187]. In-situ/Operando: Increasing interest for assessing interfacial processes in SSBs [188–190]. Approach is not limited to analysis of chemical composition/structure of interphases by a combination of TEM/STEM and EELS, SAED, or annular bright-field and high-angle annular dark-field [191–196]. Formation of SCL [197,198].
Auger Electron Spectroscopy (AES) and Scanning Auger Microscopy (SAM)	Ex-situ/In-situ/Operando	Ex-situ: Analysis/Visualization of spatial distribution of Li compounds with spectroscopic methods based on excitation by X-ray beams [160]. SAM provides lateral spatial resolution in the range of 5–100 nm [199]. Analysis of interphases in SSBs [177]. In-situ/Operando: LiNi _{0.8} Co _{0.15} Mn _{0.05} O ₂ /Li ₆ PS ₅ Cl/Li analysis [200].
Raman Spectroscopy and Microscopy	Ex-situ/In-situ/Operando	Ex-situ: Mapping study of composite positive electrode/solid electrolyte interface in LiCoO ₂ /LPS/InLi [201]. In-situ/Operando: Exploited strategy of perforated current collectors to probe buried interfaces in LiCoO ₂ Li ₆ PS ₅ Cl/Li battery [202].
X-Ray Photoelectron spectroscopy (XPS)	Ex-situ/In-situ/Operando	Ex-situ: Studied chemical evolution in interfaces in SSBs [203,204]. In-situ/Operando: Perforated current collectors can be exploited to perform operando monitoring electrochemical evolution of interfaces in SSBs [205–207].
Time-of-flight Secondary Ion Mass Spectrometry (ToF-SIMS)	Ex-situ/In-situ	Ex-situ: Analyzed buried interfaces in SSBs, similar issues found in XPS measurements [208–210]. In-situ: Evidenced partial reversible delithiation/lithiation of active material that decreased with increasing cycling from exposed interfaces [211].
X-Ray Absorption Spectroscopy (sXAS)	Ex-situ/In-situ/Operando	Ex-situ: Used as an X-ray source to illuminate sample at varied wavelengths [160].

Table 8 (continued)

Techniques	Studies	Significance
Scanning Probe Microscopy (SPM)	Ex-situ In-situ	In-situ/Operando: Used as Al current collector for either composite NMC111 or LiFePO ₄ (LFP) cathodes [212]. Ex-situ: Analyzed interfaces in SSBs. Used for assessing interfacial phenomena [213]. In-situ/Operando: Tracked potential drops created in SSBs after electrochemical polarisation [214, 215].

Table 9
Common Bulk-Approach techniques.

Techniques	Studies	Significance
Neutron Depth Profiling (NDP)	In-situ Operando	In-situ/Operando: Tracked diffusion of Li between cathode and anode in LiCoO ₂ (LCO)/LiPON/Cu micro battery [216]. Low energy neutron (n) interacts with a ⁶ Li atom producing an α (⁴ He) and a triton (³ H) particle [160]. ⁶ Li + n → ⁴ He + ³ H
X-Ray Computed Tomography (X-CT)	Ex-situ/In-situ/ Operando	Ex-situ: Enabling the visualization and evaluation of the microstructure of materials in SSBs [160]. Provides analysis for electrochemical performance and stability of device [217]. Research on symmetric Li/solid electrolyte/Li cells [160]. In-situ/Operando: Assess the evolution of microstructure of solid electrolyte in Li/Li _{1-x} Al _x Ge _{2-x} (PO ₄) ₃ (LAGP)/Li symmetric cell [170].
Nuclear Magnetic Resonance (NMR)	Ex-situ	Ex-situ: Assessed diffusion and distribution of Li ⁺ and Li metal at interfaces [160]. Applications to energy storage devices [218]. Interfacial studies on SSBs [160]. Tracked formation and growth of Li dendrites by ⁷ Li magnetic resonance imaging (MRI) [173].

6.3. Optimization of fast charging capabilities

The advancement of fast-charging technologies is a pivotal aspect of the commercial viability and consumer adoption of SSBs, particularly within the automotive sector. This requires a concerted effort to enhance the batteries' intrinsic fast charging capabilities and integrate sophisticated power management systems. These developments are crucial for meeting consumer expectations and safety standards while extending the operational longevity of the batteries. The need to improve the ionic conductivity and electron mobility within the battery is at the core of optimizing fast charging. Research should focus on material modifications, such as doping or creating nanostructured materials that facilitate rapid ion transport while maintaining structural stability under rapid charge and discharge cycles. Additionally, reducing the battery's internal resistance through improved electrode and solid electrolyte interfaces can significantly enhance charge rates.

Addressing thermal management is critical in fast-charging systems. High rates of charging can lead to significant heat generation, which, if not properly managed, can deteriorate battery health and safety. Implementing advanced thermal management solutions, such as heat sinks, cooling systems, or phase change materials, will be essential to dissipate heat efficiently and maintain the battery's temperature within safe operational limits. Overall, integrating advanced power

management systems is essential for optimizing the charging process and ensuring battery longevity. These systems should be capable of dynamically adjusting the charging rates based on real-time assessments of battery conditions and external factors such as ambient temperature and battery age. Smart charging algorithms that can modulate charge rates to minimize stress and degradation of the battery cells will play a critical role in extending the life span of SSBs.

Fast charging involves higher currents and voltages that increase the risk of mechanical and chemical instabilities within the battery. Developing robust safety mechanisms and protocols is imperative to prevent overcharging, thermal runaway, and other potentially hazardous conditions. These include incorporating sensors that provide continuous monitoring and diagnostics, coupled with automated cutoff systems that can intervene in real time to prevent failures. The optimization of SSB fast charging capabilities also involves ensuring compatibility with existing and future electric vehicle charging infrastructure. Collaborations between battery developers, vehicle manufacturers, and infrastructure providers will be crucial to harmonize the technical specifications and ensure seamless integration into the broader EV ecosystem.

By addressing these key areas, the research and development community can enhance the fast-charging capabilities of SSBs, thus meeting the stringent requirements of the automotive industry and other sectors demanding quick energy replenishment. This will not only boost consumer confidence in electric vehicles but also significantly contribute to the broader adoption of clean energy technologies.

6.4. Enhanced In-Operando characterization techniques

The evolution of in-operando characterization methods is crucial for the detailed, real-time monitoring and proactive management of the health and performance of solid-state batteries (SSBs). These advanced methodologies are instrumental in ensuring optimal operational parameters, enhancing safety profiles, and prolonging the lifespan of these energy storage systems. Developing sophisticated sensing and diagnostic technologies is essential to implementing precise control strategies that adapt to changing operational conditions and maintain battery integrity throughout its lifecycle. At the forefront of these efforts are the enhancements in sensor technology capable of providing real-time data on critical battery parameters such as temperature, voltage, current, and internal resistance. Innovations may include the integration of nano-sensors within the battery structure that can detect microstructural changes at the electrode and electrolyte levels. These sensors enable the early detection of potential failures, such as electrolyte decomposition or electrode delamination, before they escalate into serious problems. The integration of high-resolution diagnostic tools, such as acoustic microscopy, optical coherence tomography, and advanced spectroscopy techniques, can offer unprecedented insights into the internal processes of SSBs. These tools allow for the visualization and analysis of the dynamic interactions within the battery during charge and discharge cycles, providing valuable data to optimize battery chemistry and architecture.

Another critical area is leveraging data analytics and machine learning algorithms to interpret the vast amounts of data generated by in-operando diagnostics. These computational tools can predict battery behavior under various conditions and identify optimal operating parameters. By analyzing trends and patterns in the data, machine learning models can forecast potential system failures and suggest preventive measures to extend battery life and enhance performance. Developing dynamic feedback mechanisms that adjust battery operation based on real-time diagnostics is essential for maintaining battery health. These systems use data from in-operando characterization to modulate charging strategies, manage power output, and activate safety protocols when anomalies are detected. Such adaptive management not only optimizes performance but also significantly mitigates risks associated with battery degradation and failure.

It will be crucial to standardize these advanced in-operando characterization techniques and validate their accuracy and reliability across different battery technologies and applications. This ensures compatibility and comparability of data, facilitating broader implementation and acceptance of these techniques in quality control and regulatory frameworks. By expanding the capabilities of in-operando characterization techniques, researchers and engineers can significantly enhance the control strategies for SSBs, thereby maximizing their efficiency, safety, and longevity.

7. Conclusion

The advancements presented in this review will provide the necessary technological foundation to support the deployment of SSBs across a wide range of applications, from consumer electronics to electric vehicles, aligning with the demands of modern energy storage solutions. As SSBs become more prevalent, establishing comprehensive safety standards and regulatory guidelines will be crucial. These frameworks should ensure the safe application of SSBs across various domains, including consumer electronics and electric mobility. In closing, addressing the key pivotal points discussed will enable the research community and industry stakeholders to advance SSB technology towards significant market penetration and scalable manufacturing, fulfilling its promise as a cornerstone of next-generation energy storage systems.

CRediT authorship contribution statement

Thanh Vy Nguyen: Writing – review & editing. **Talyah Nelson:** Writing – review & editing. **Ruoxu Shang:** Writing – review & editing. **Cengiz S. Ozkan:** Writing – review & editing. **ozkan mihrimah:** Writing – review & editing. **Brian Abaoag:** Writing – review & editing. **Harsha Antony:** Writing – review & editing. **Charlotte Nelson:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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